

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

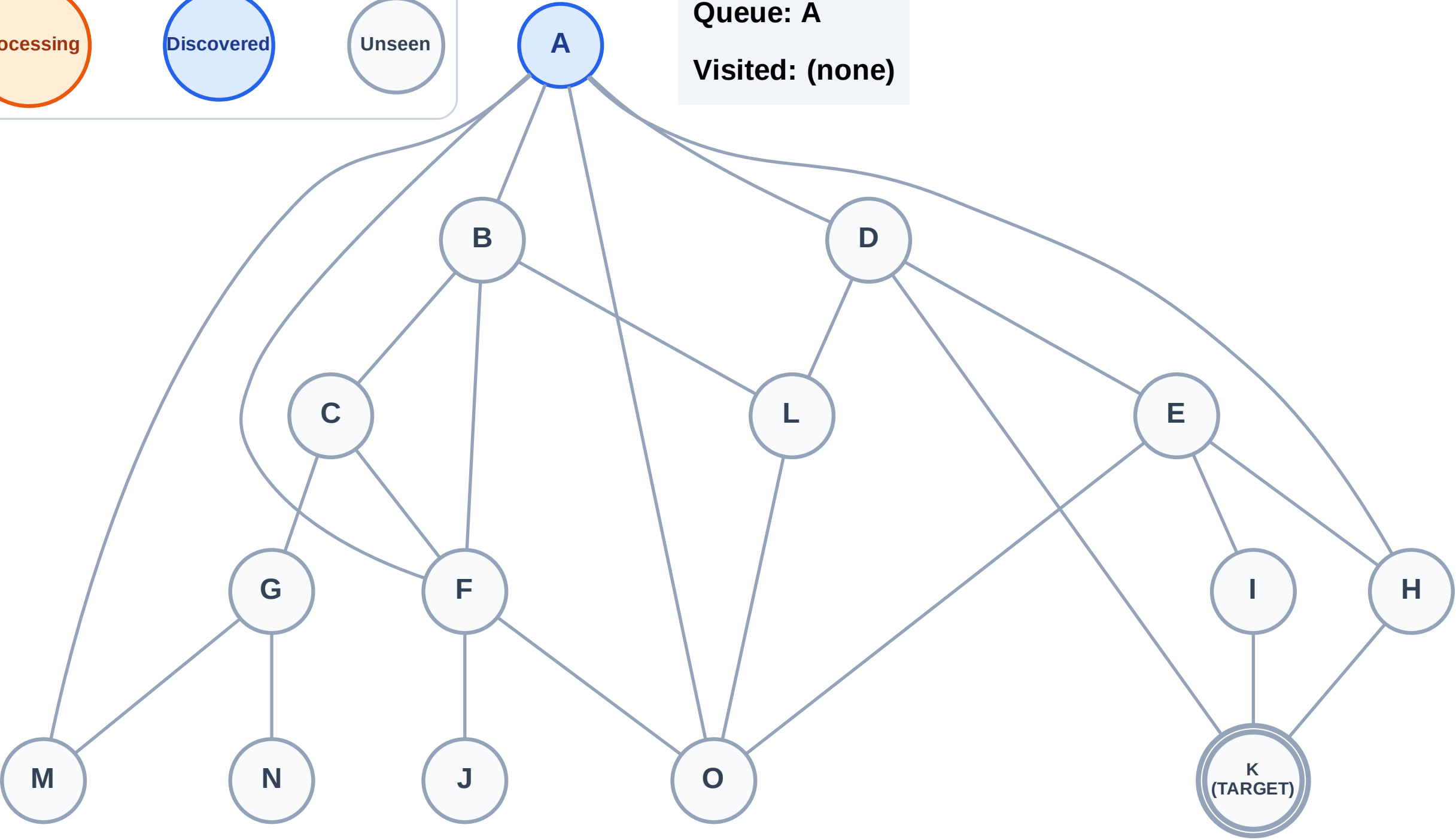
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

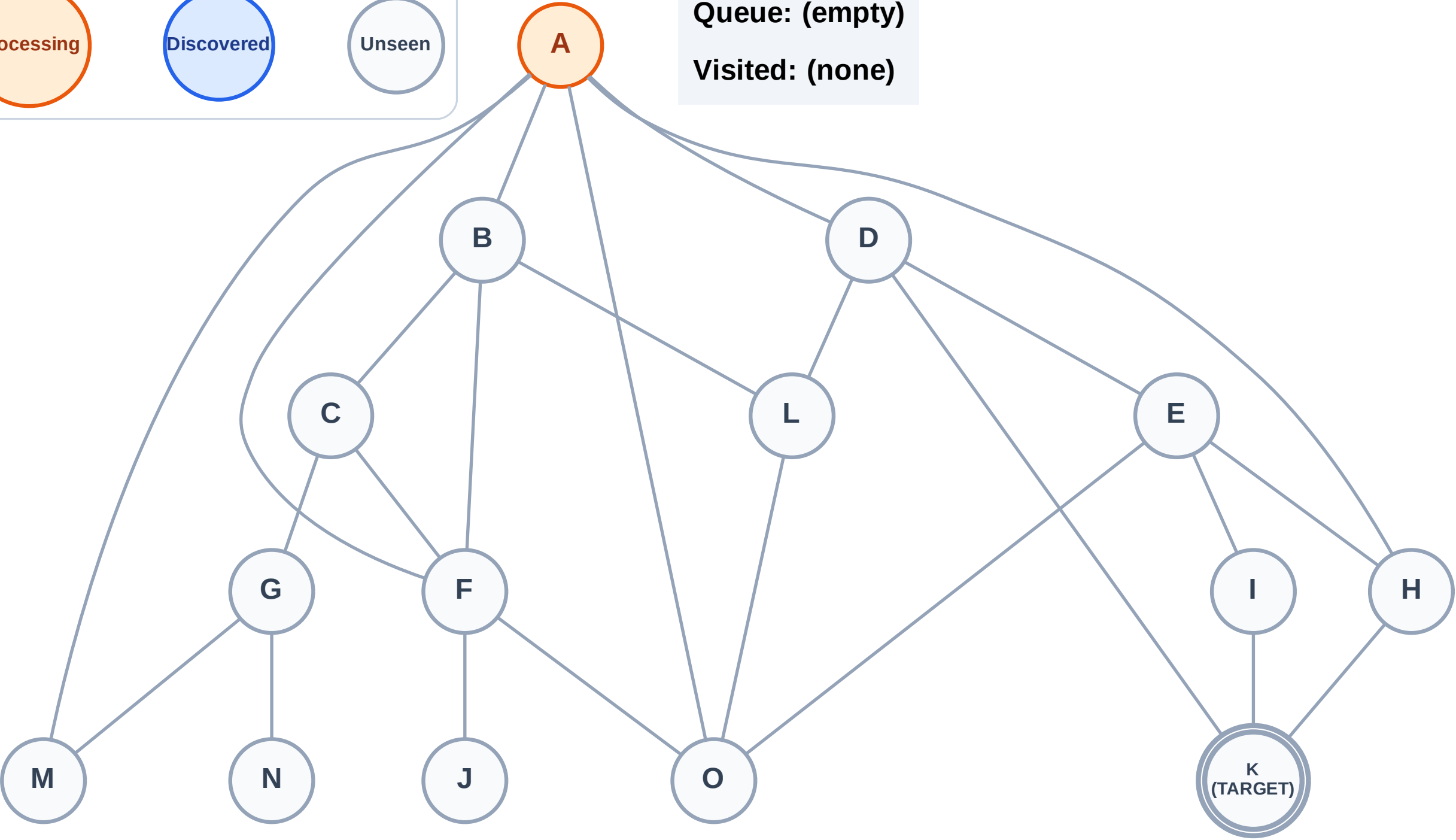
Complete

Processing

Discovered

Unseen

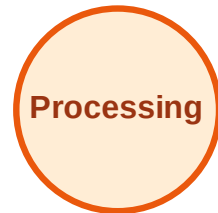
Queue: (empty)
Visited: (none)



BFS Traversal

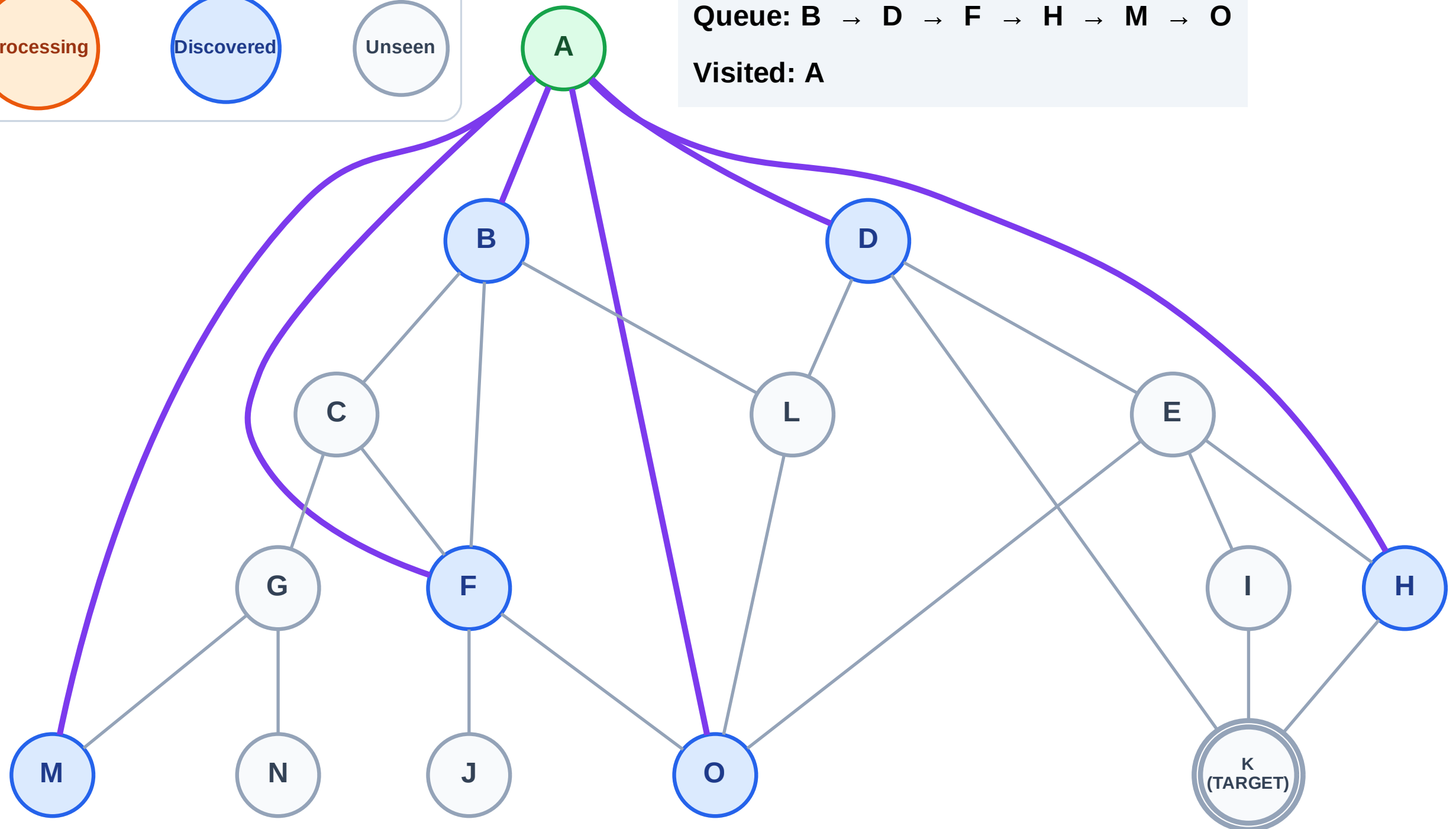
Step 3: Discover B, D, F, H, M, O from A

Legend

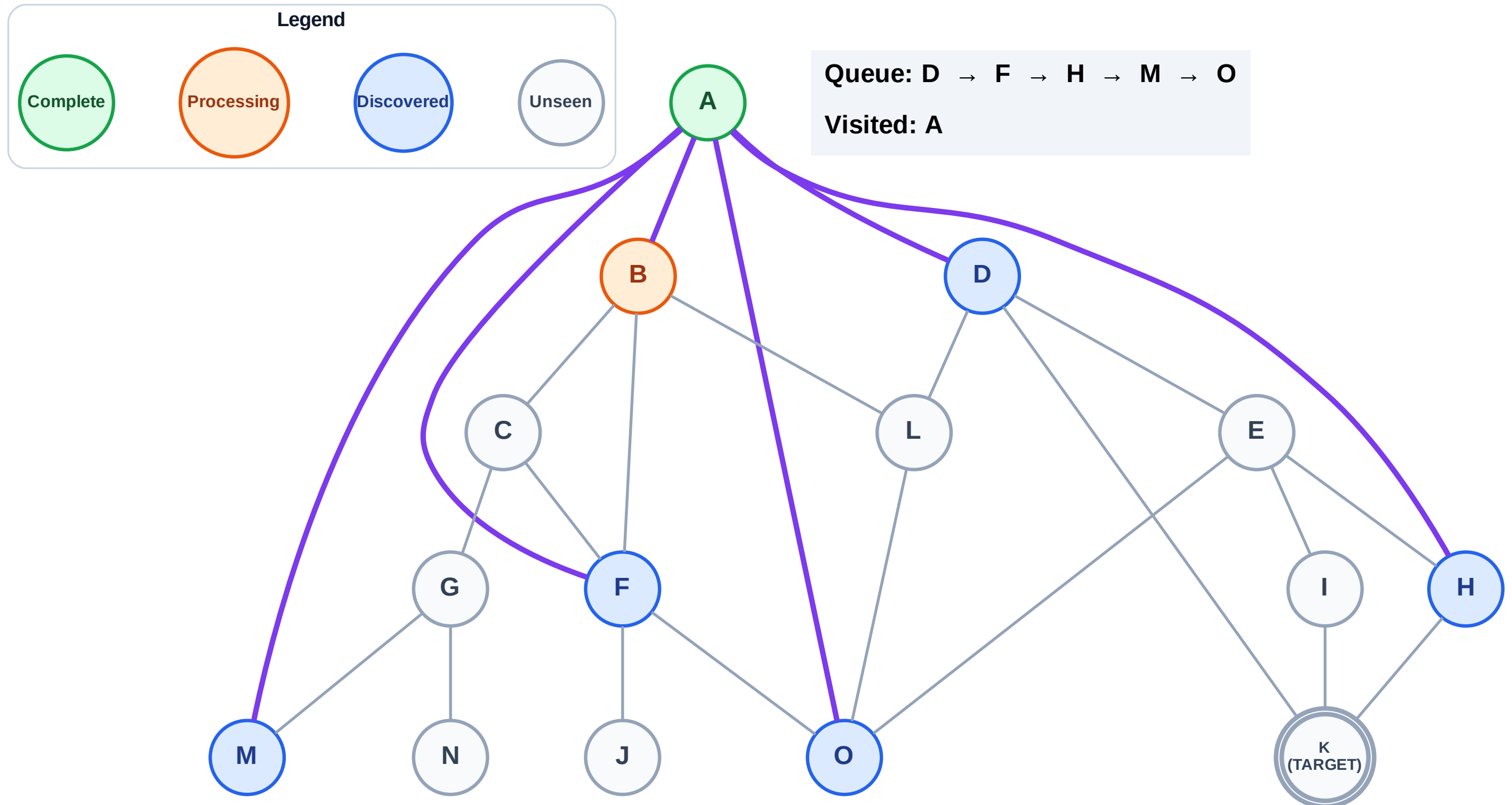


Queue: B → D → F → H → M → O

Visited: A



Step 4: Process node B



BFS Traversal

Step 5: Discover C, L from B

Legend

Complete

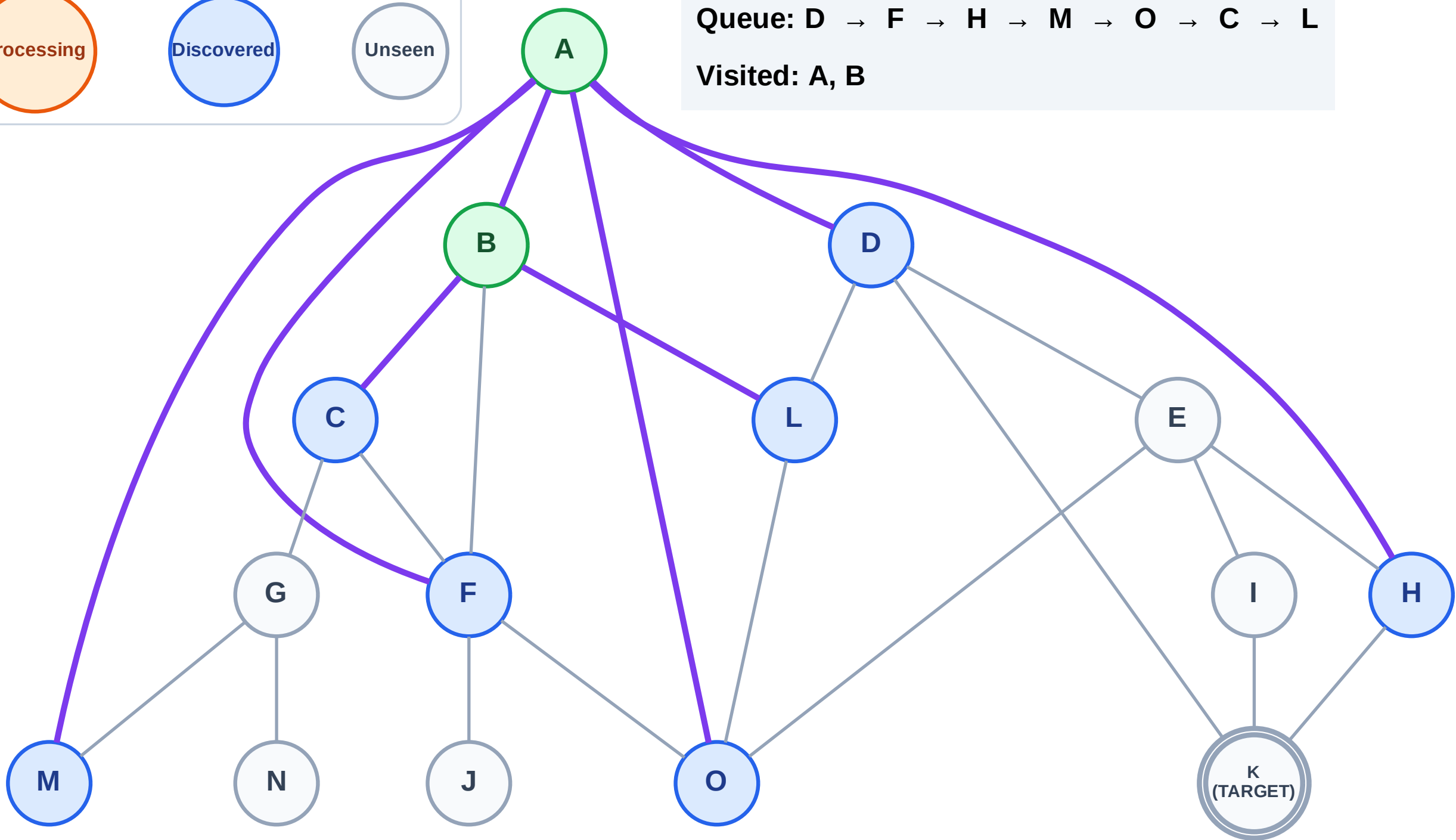
Processing

Discovered

Unseen

Queue: D → F → H → M → O → C → L

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

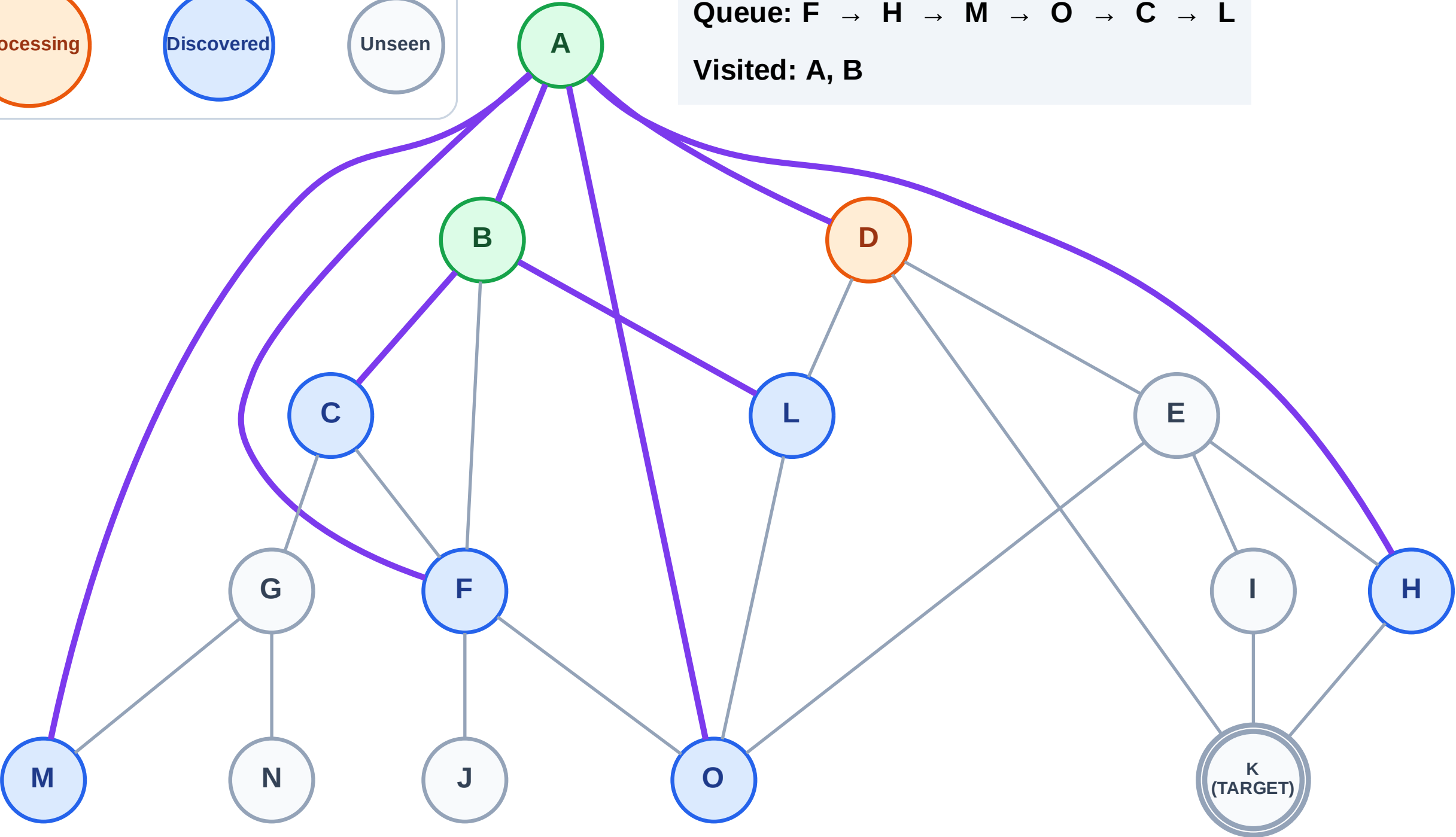
Processing

Discovered

Unseen

Queue: F → H → M → O → C → L

Visited: A, B



BFS Traversal

Step 7: Discover E, K from D

Legend

Complete

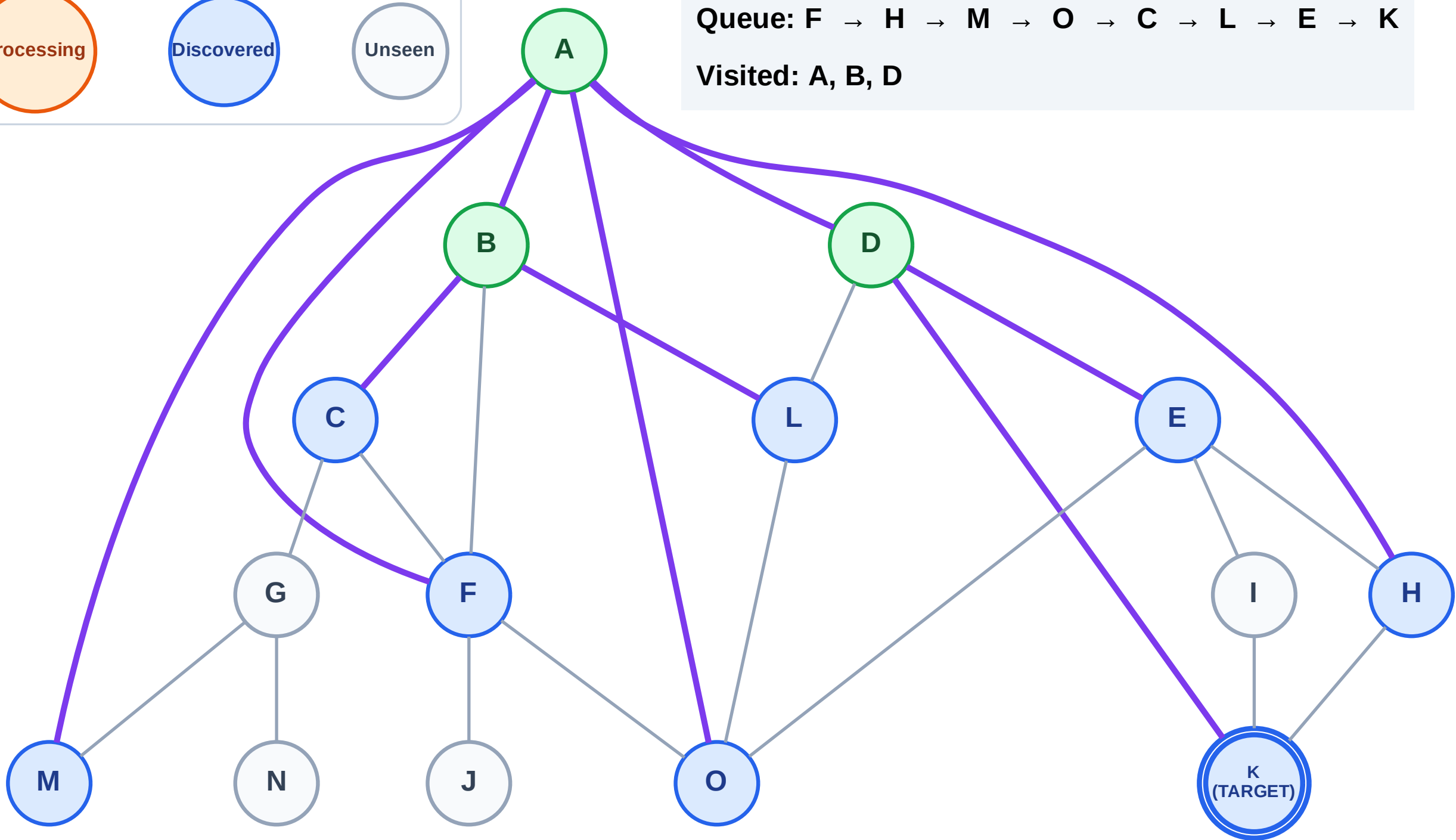
Processing

Discovered

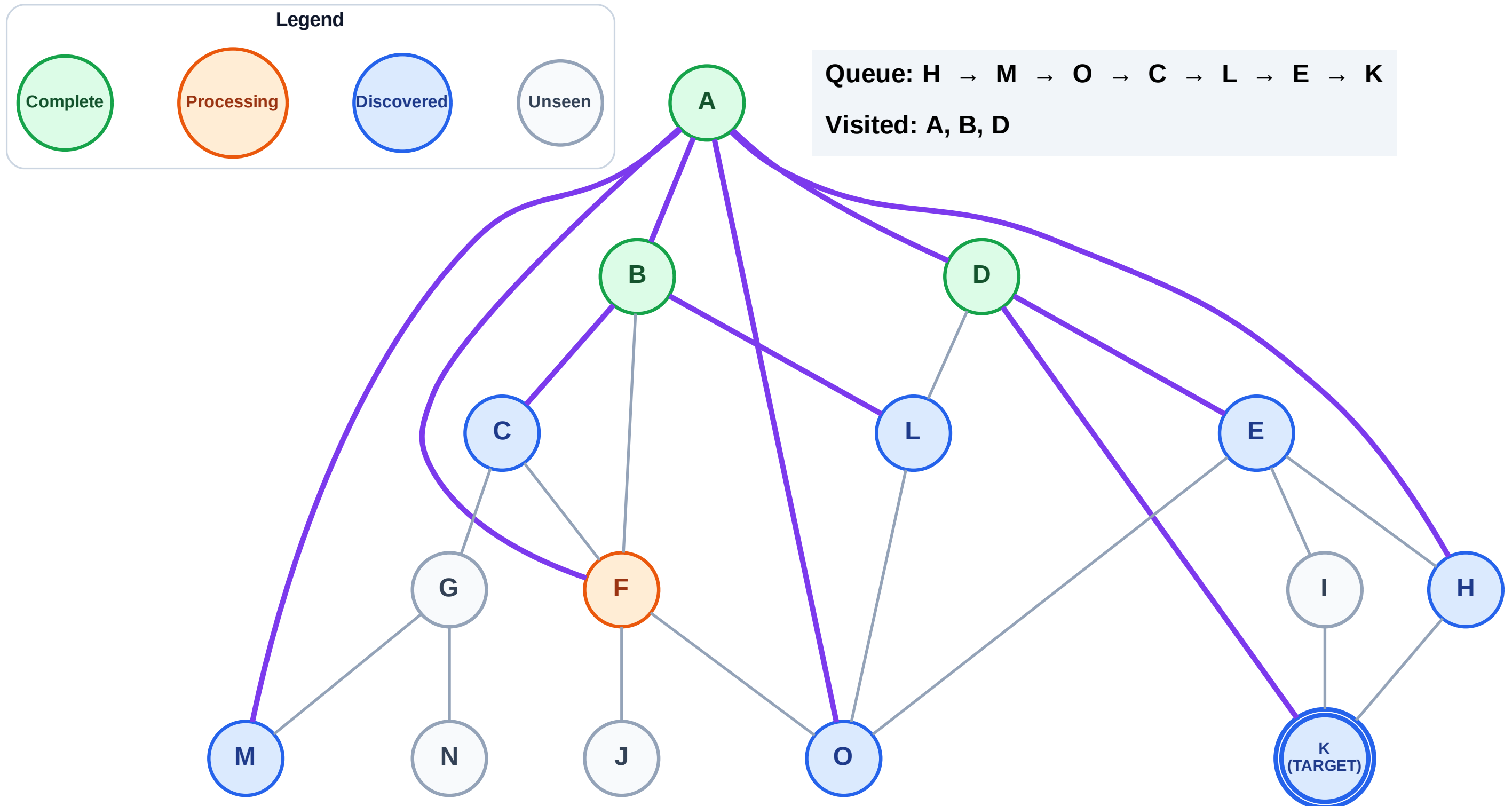
Unseen

Queue: F → H → M → O → C → L → E → K

Visited: A, B, D



Step 8: Process node F



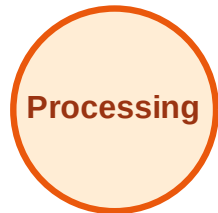
BFS Traversal

Step 9: Discover J from F

Legend



Complete



Processing



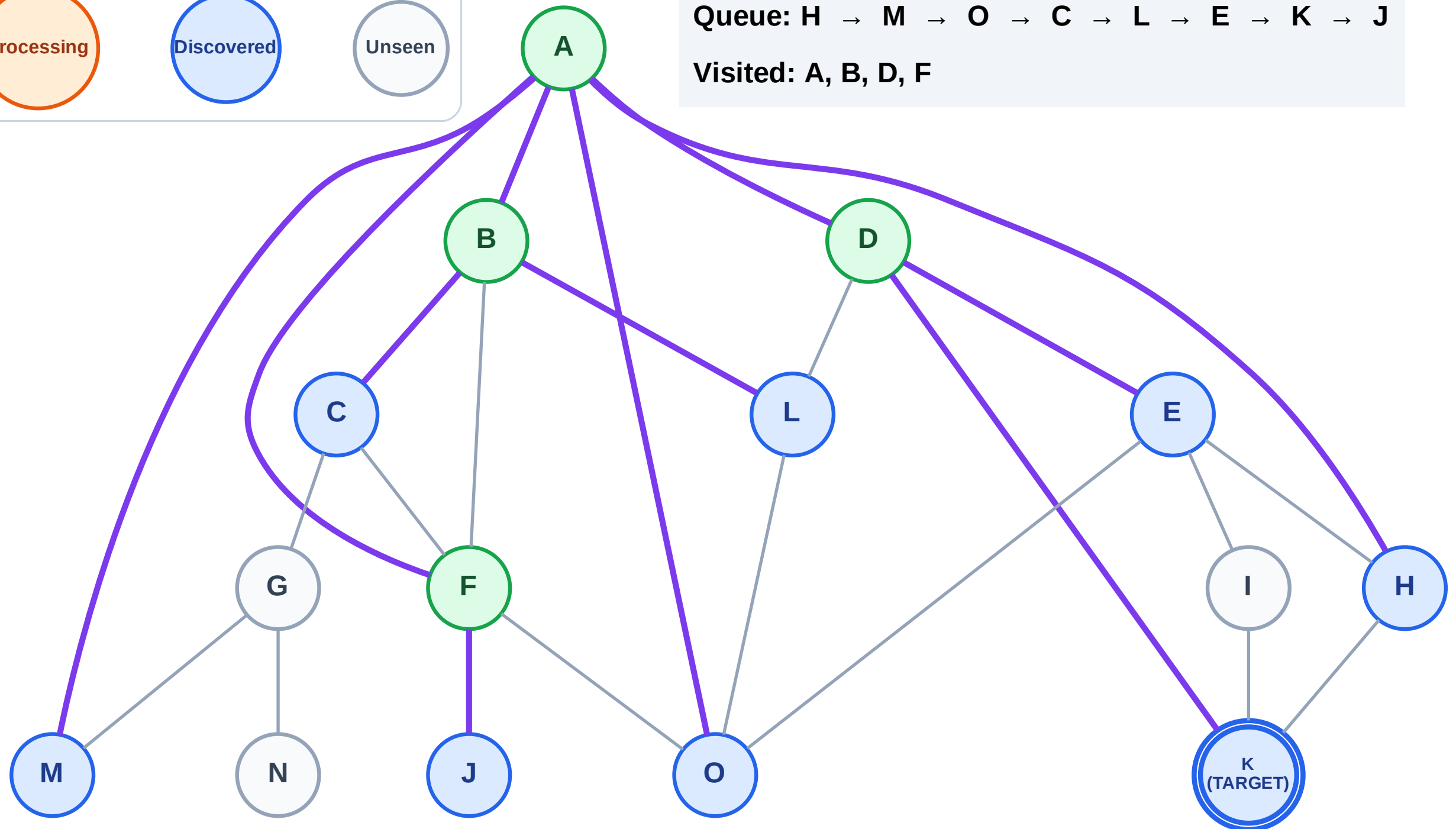
Discovered



Unseen

Queue: H → M → O → C → L → E → K → J

Visited: A, B, D, F



BFS Traversal

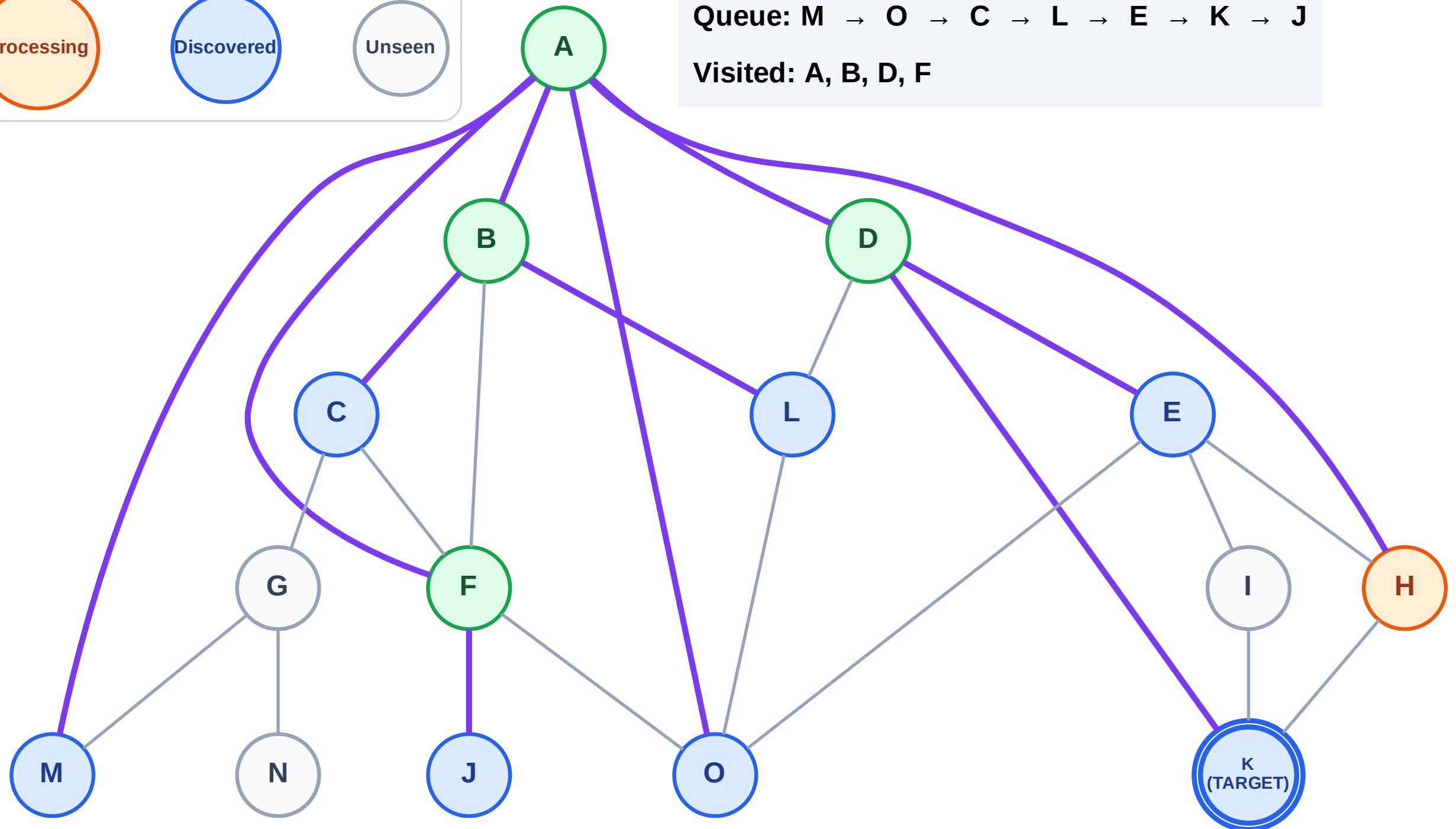
Step 10: Process node H

Legend



Queue: M → O → C → L → E → K → J

Visited: A, B, D, F



BFS Traversal

Step 11: No new neighbors from H

Legend

Complete

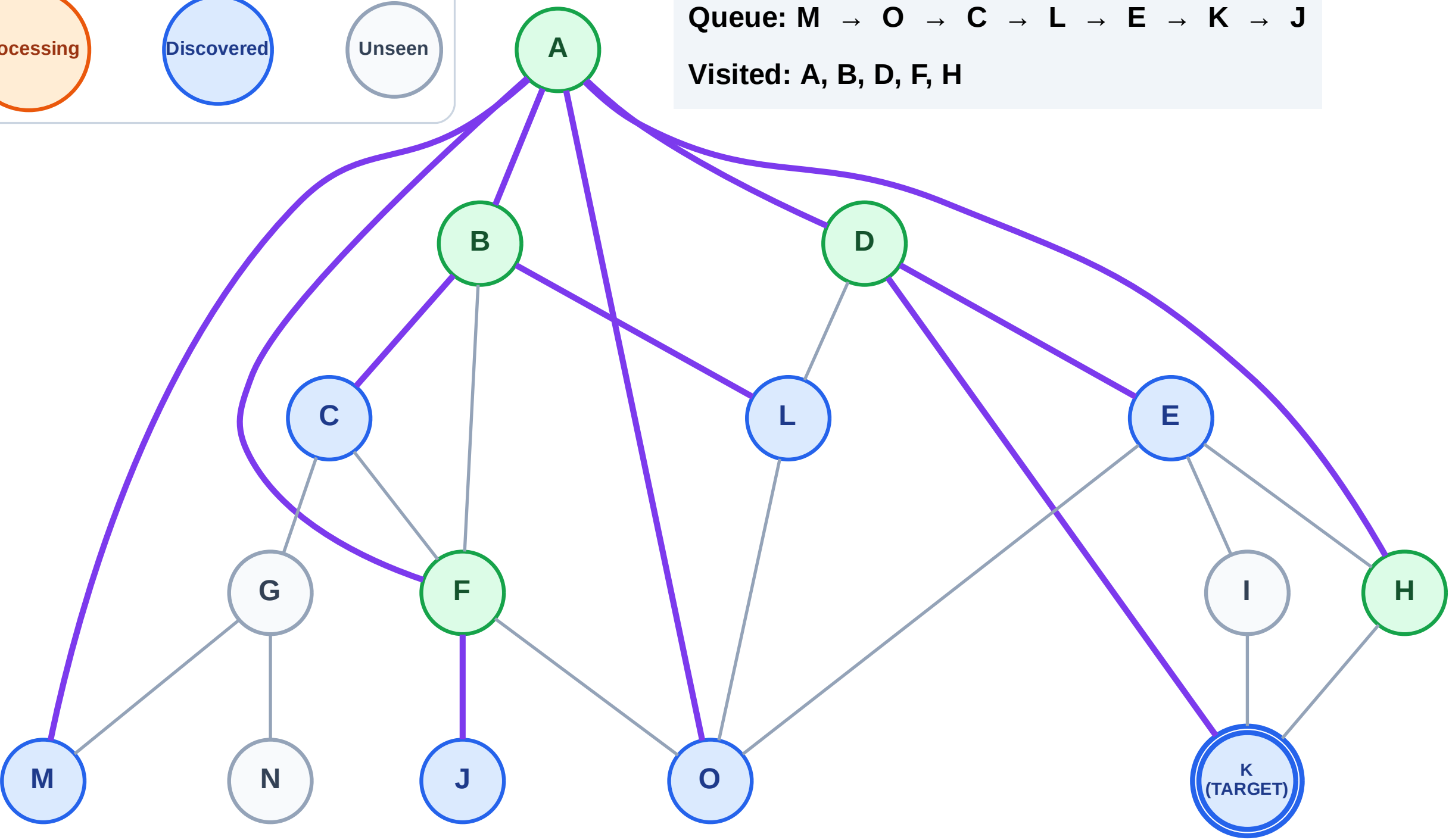
Processing

Discovered

Unseen

Queue: M → O → C → L → E → K → J

Visited: A, B, D, F, H



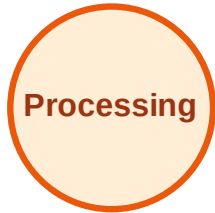
BFS Traversal

Step 12: Process node M

Legend



Complete



Processing



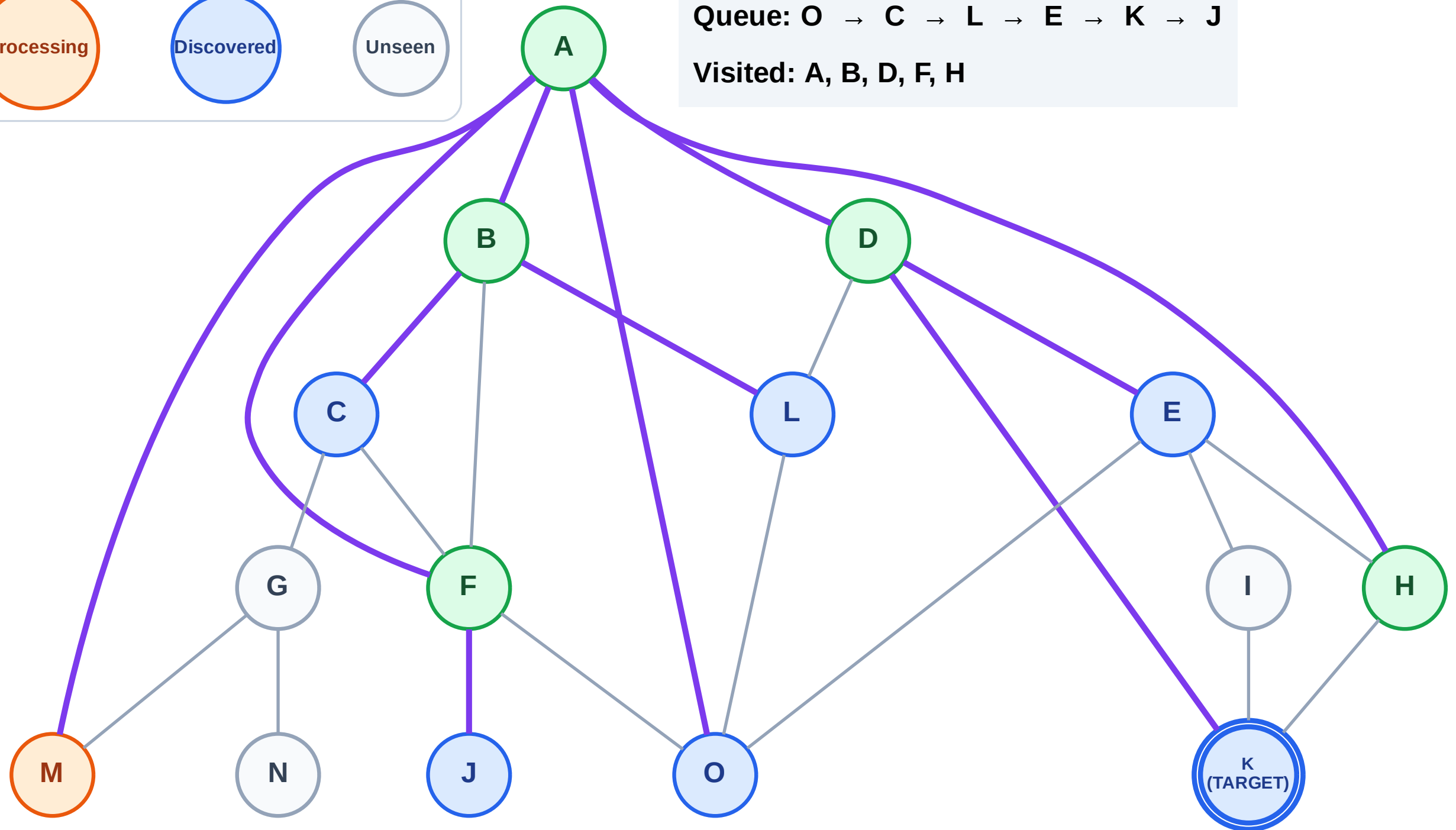
Discovered



Unseen

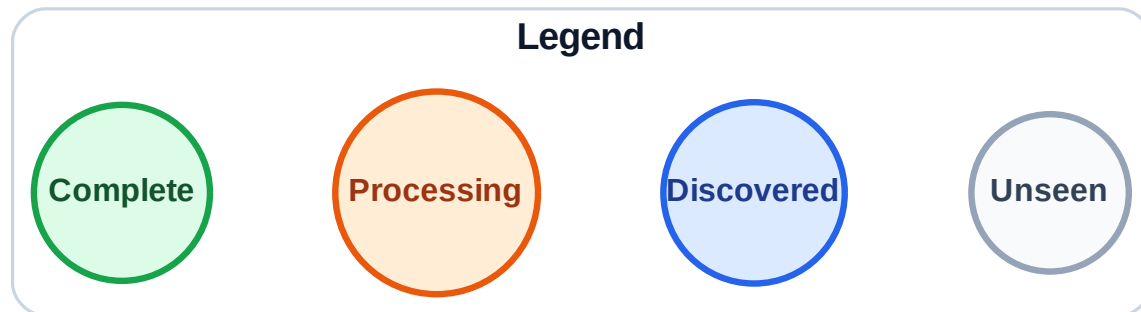
Queue: O → C → L → E → K → J

Visited: A, B, D, F, H



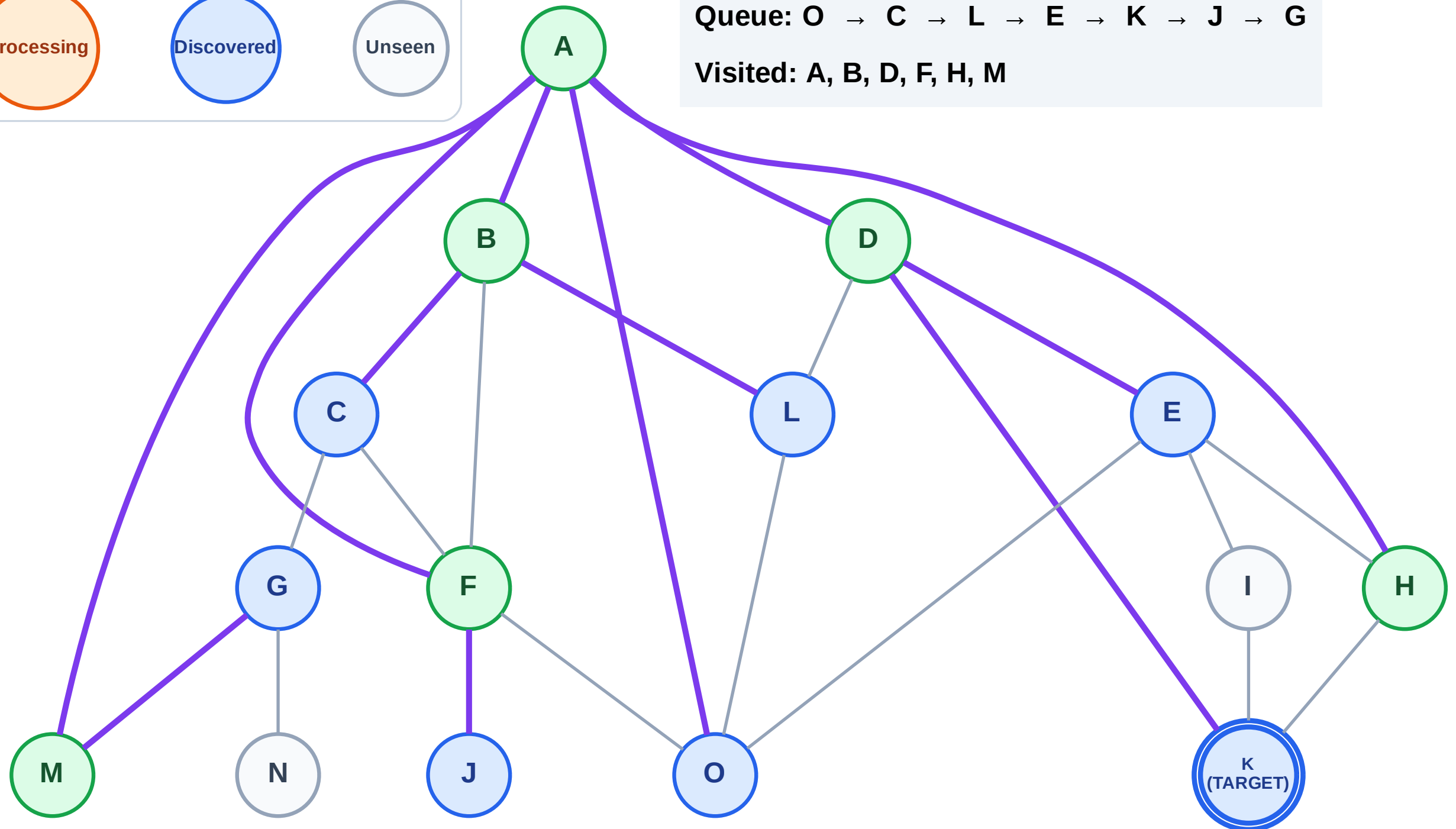
BFS Traversal

Step 13: Discover G from M



Queue: O → C → L → E → K → J → G

Visited: A, B, D, F, H, M



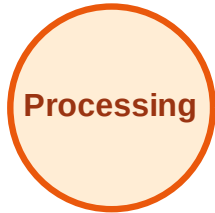
BFS Traversal

Step 14: Process node O

Legend



Complete



Processing



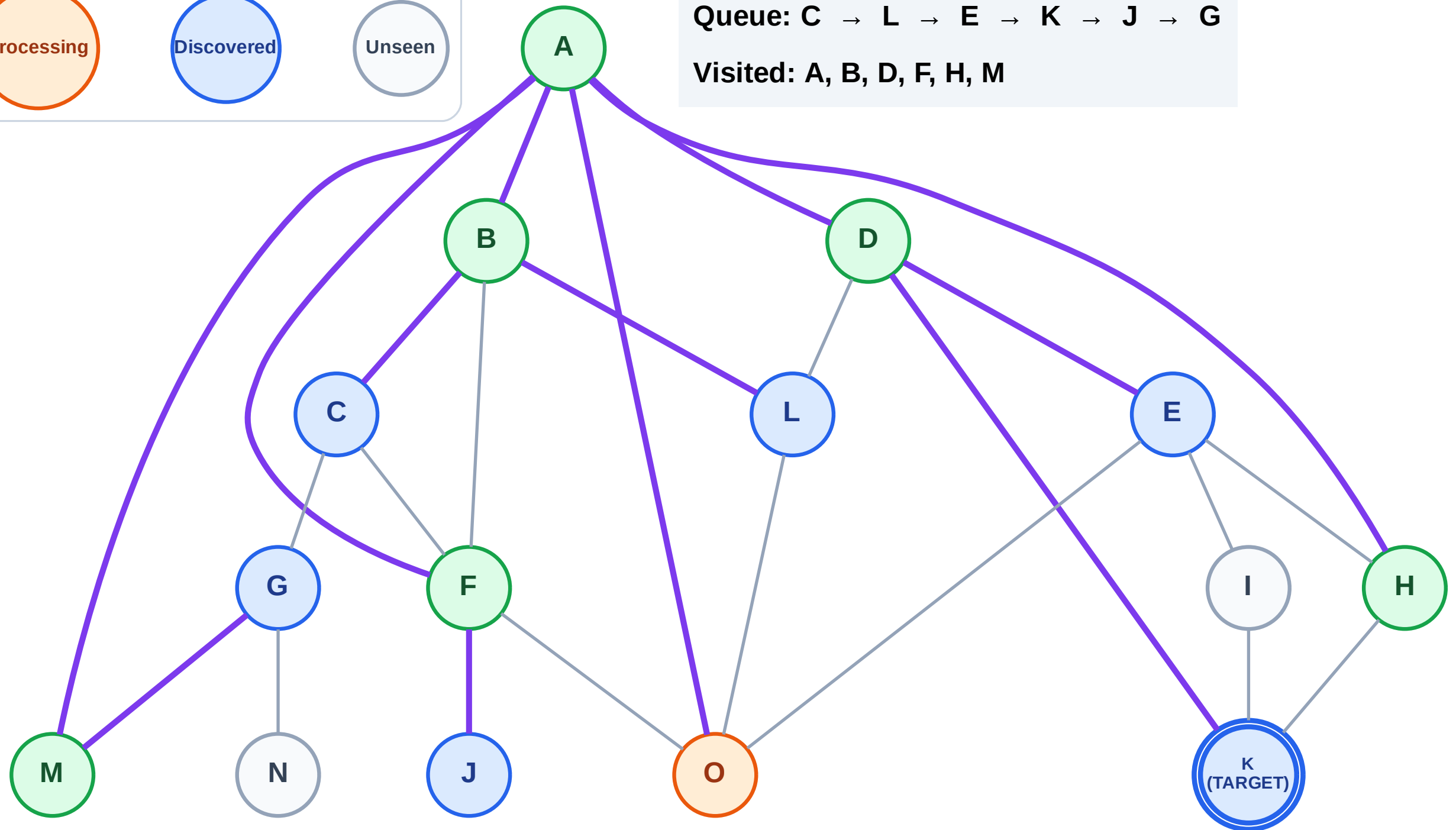
Discovered



Unseen

Queue: C → L → E → K → J → G

Visited: A, B, D, F, H, M



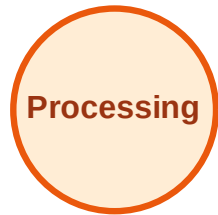
BFS Traversal

Step 15: No new neighbors from O

Legend



Complete



Processing



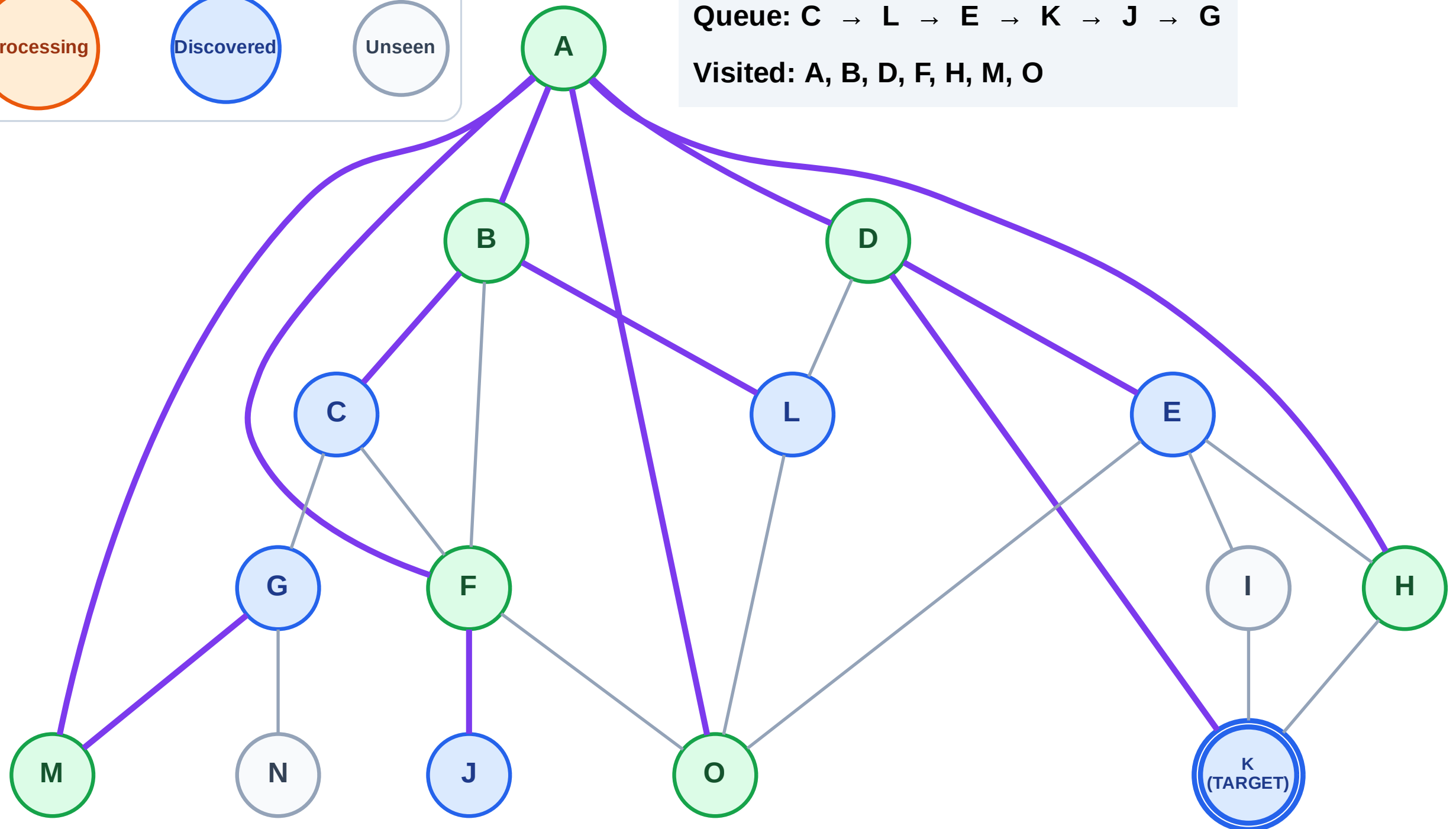
Discovered



Unseen

Queue: C → L → E → K → J → G

Visited: A, B, D, F, H, M, O



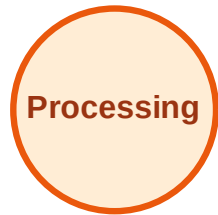
BFS Traversal

Step 16: Process node C

Legend



Complete



Processing



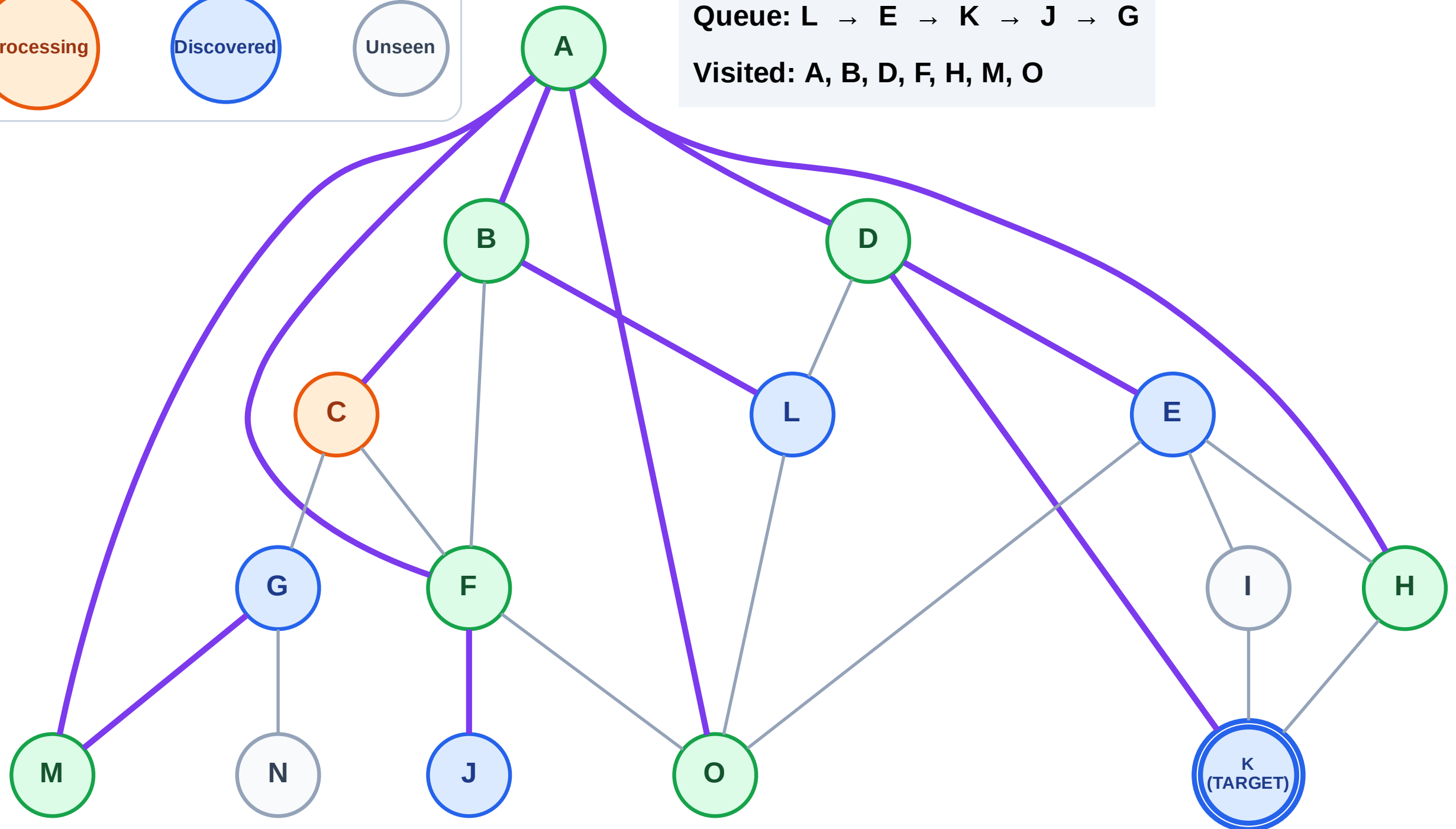
Discovered



Unseen

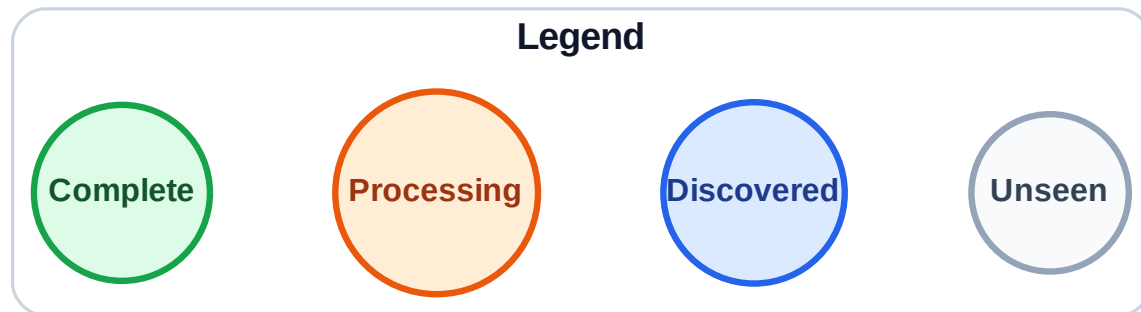
Queue: L → E → K → J → G

Visited: A, B, D, F, H, M, O



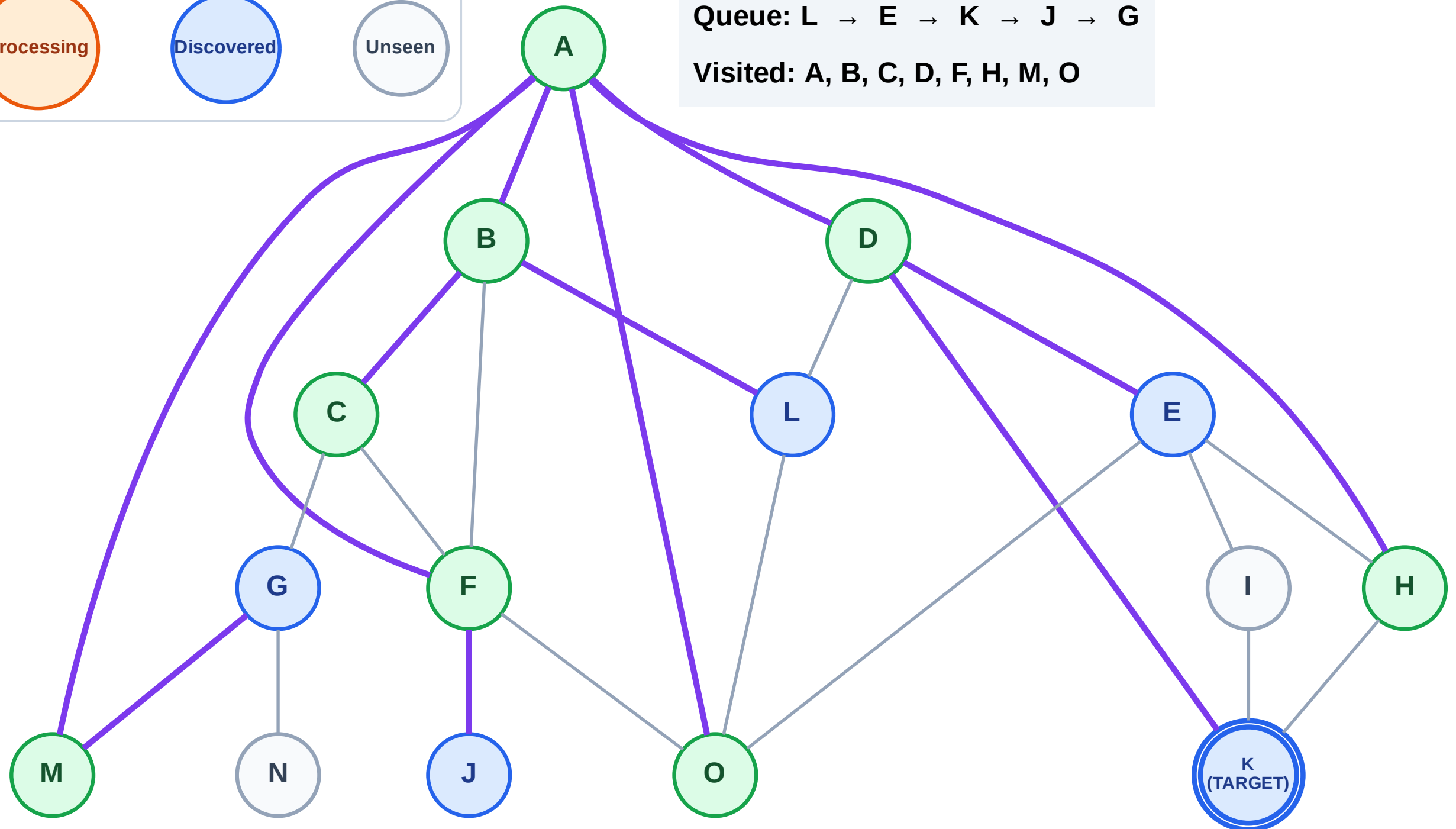
BFS Traversal

Step 17: No new neighbors from C



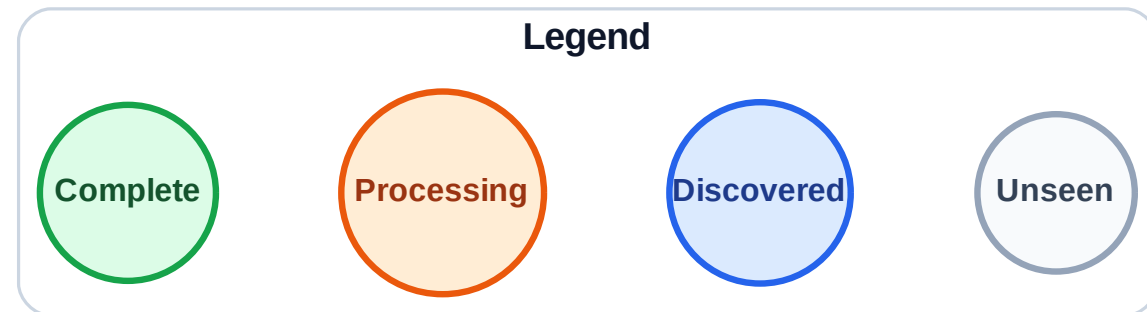
Queue: L → E → K → J → G

Visited: A, B, C, D, F, H, M, O



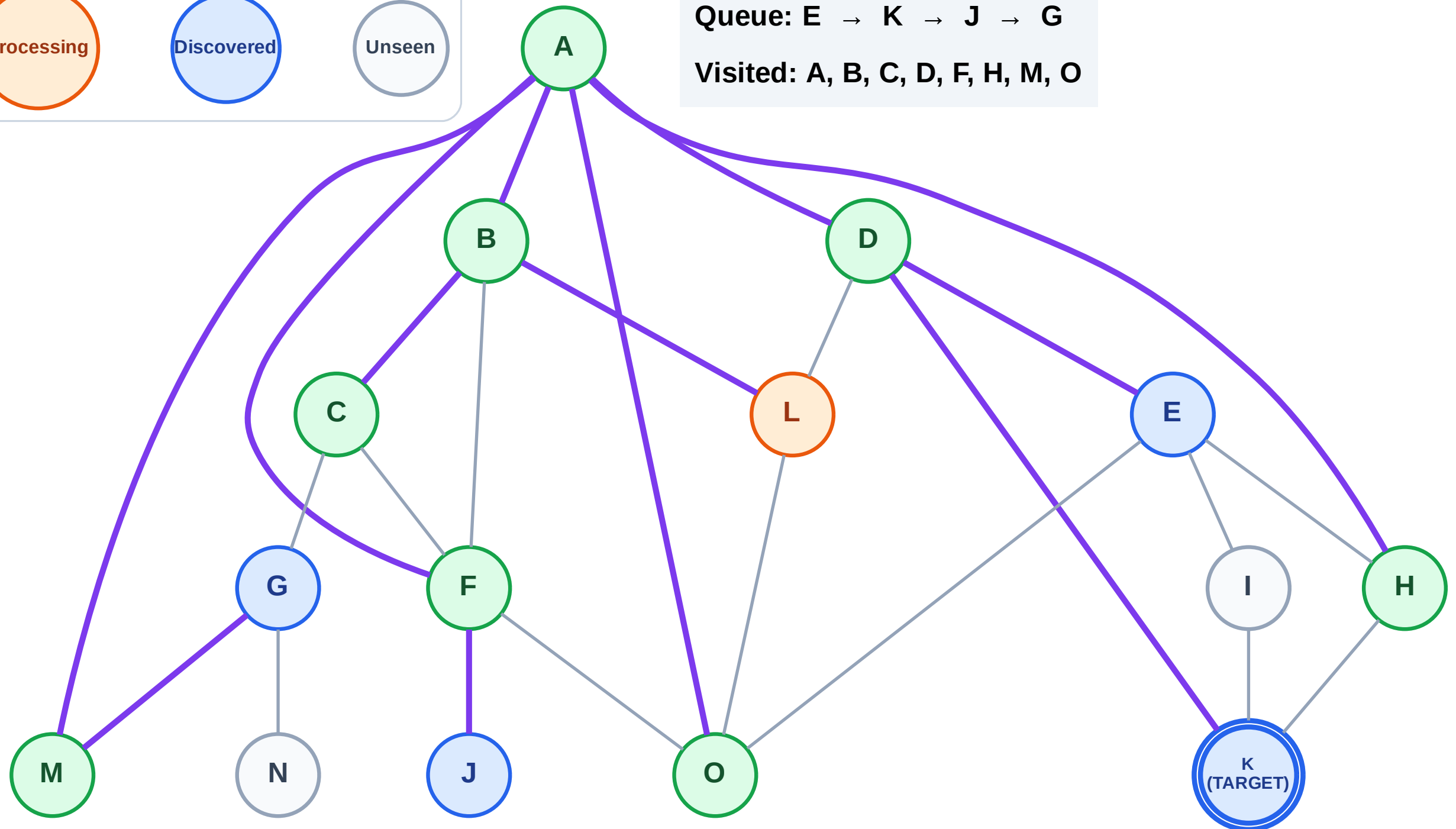
BFS Traversal

Step 18: Process node L



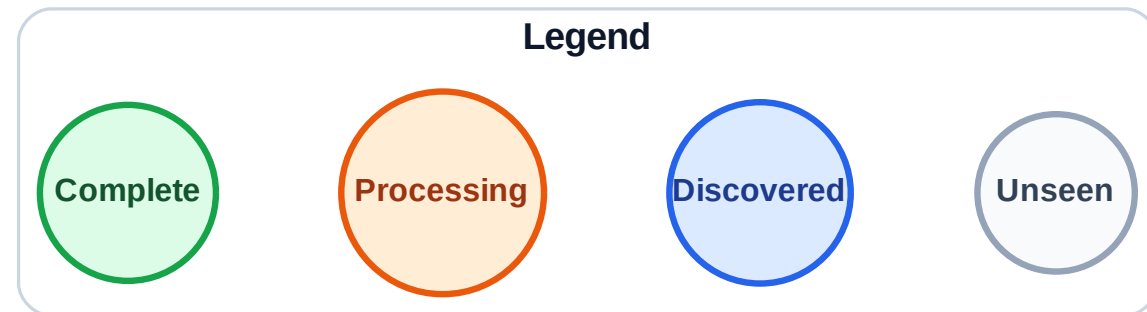
Queue: E → K → J → G

Visited: A, B, C, D, F, H, M, O



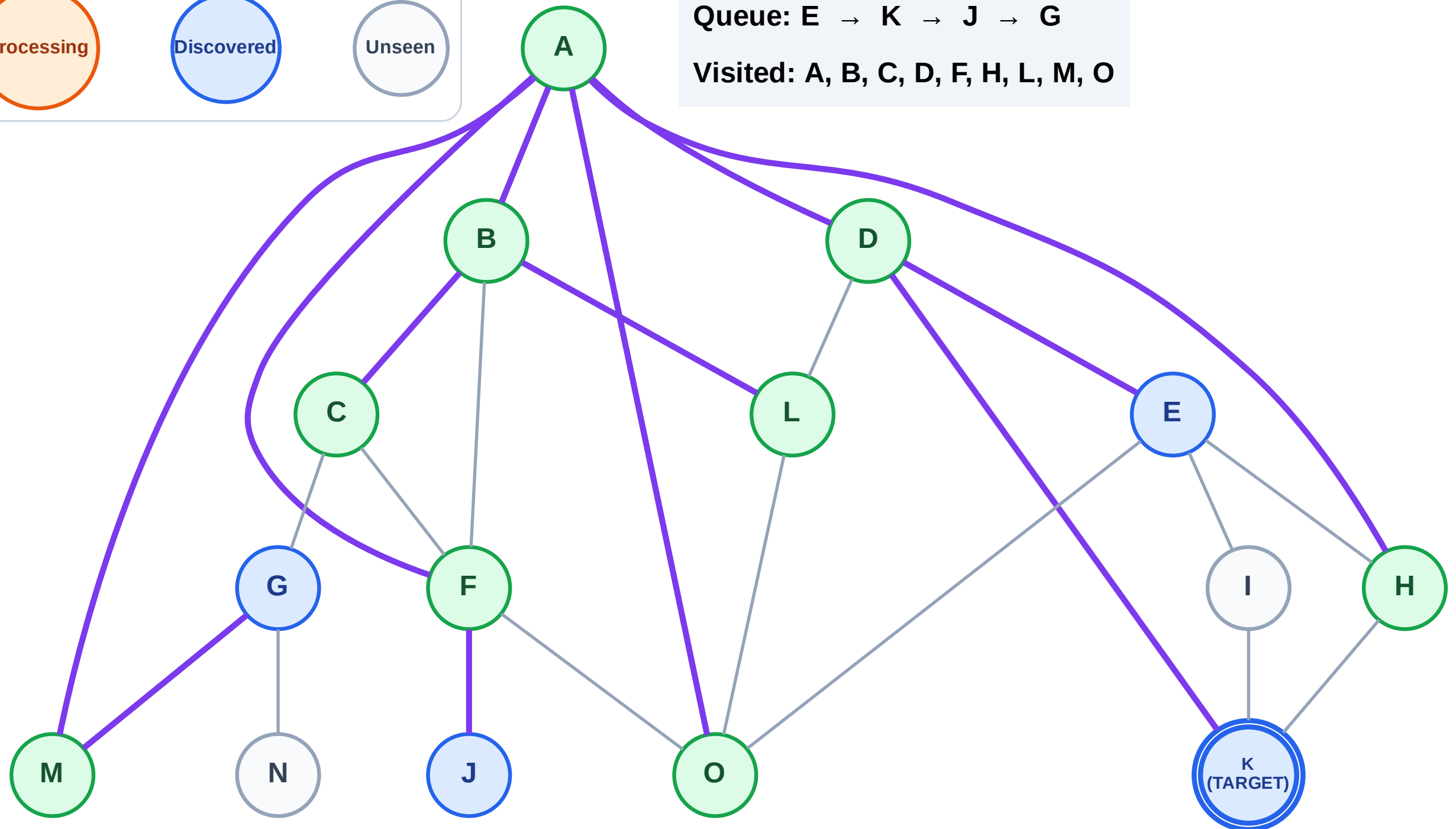
BFS Traversal

Step 19: No new neighbors from L



Queue: E → K → J → G

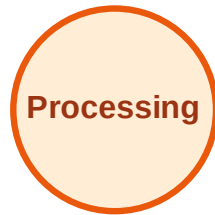
Visited: A, B, C, D, F, H, L, M, O



BFS Traversal

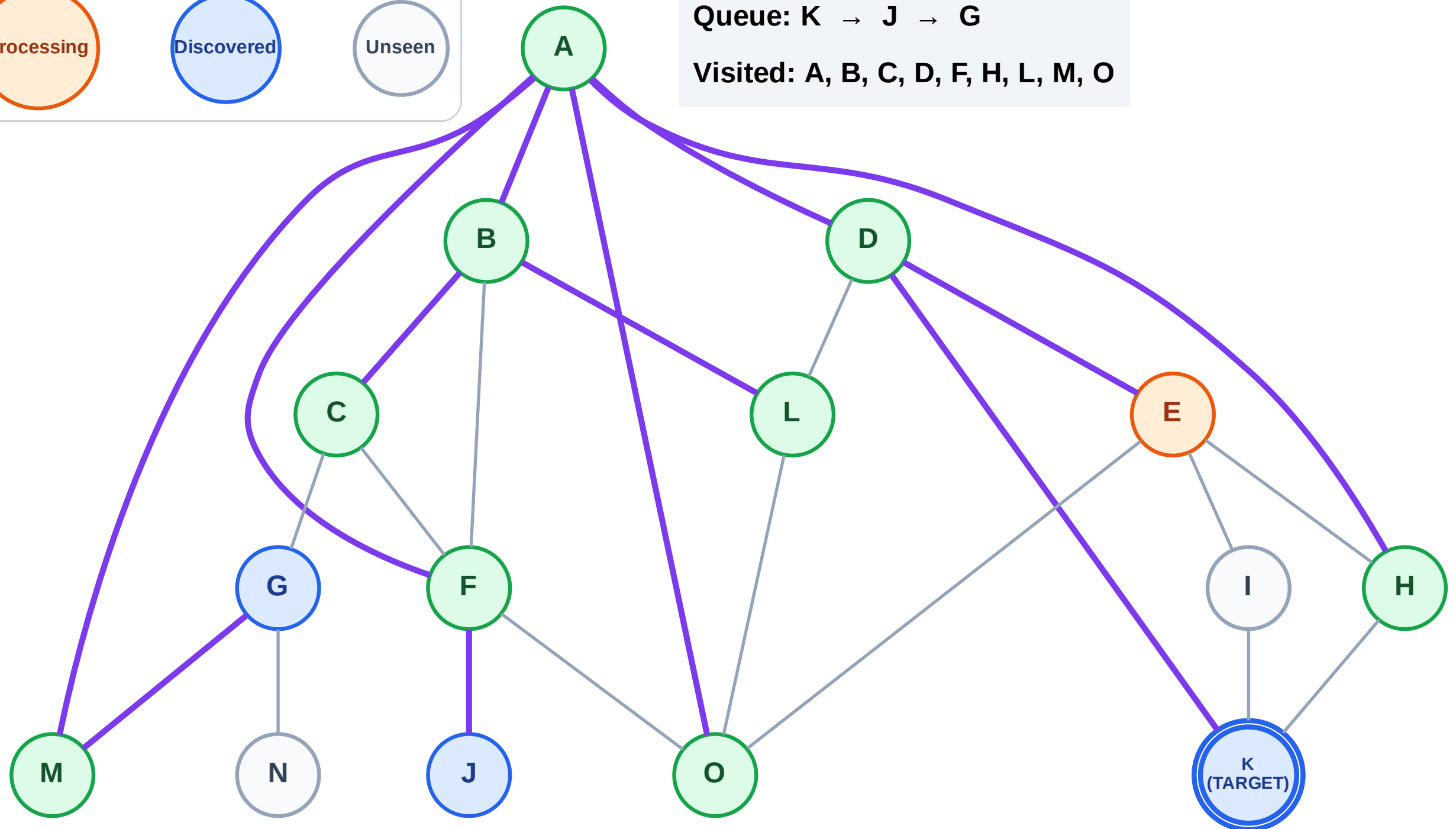
Step 20: Process node E

Legend



Queue: K → J → G

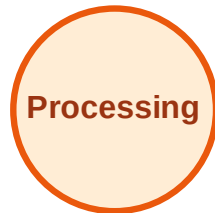
Visited: A, B, C, D, F, H, L, M, O



BFS Traversal

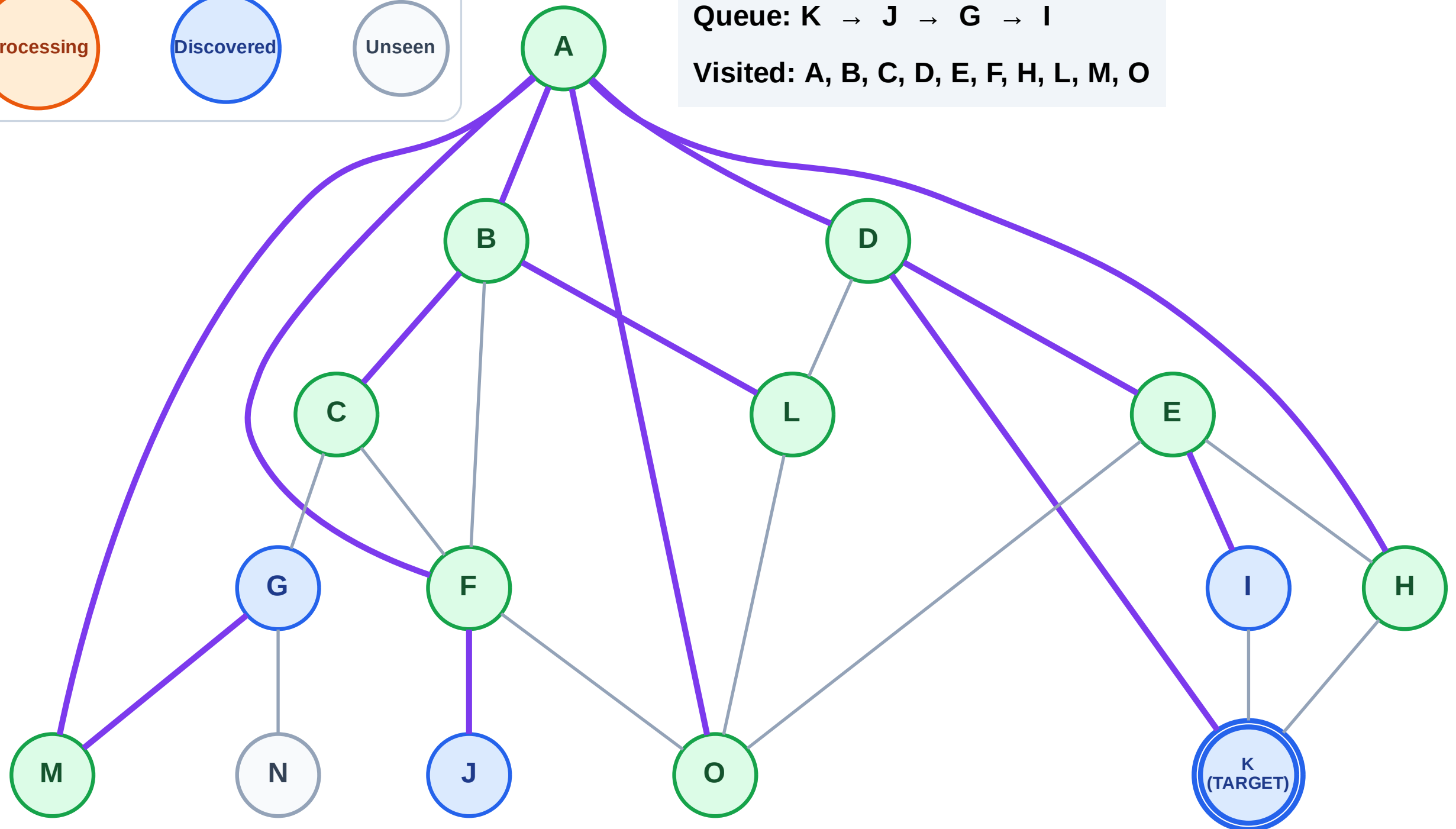
Step 21: Discover I from E

Legend



Queue: K → J → G → I

Visited: A, B, C, D, E, F, H, L, M, O



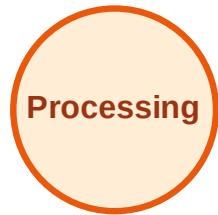
BFS Traversal

Step 22: Process node K

Legend



Complete



Processing



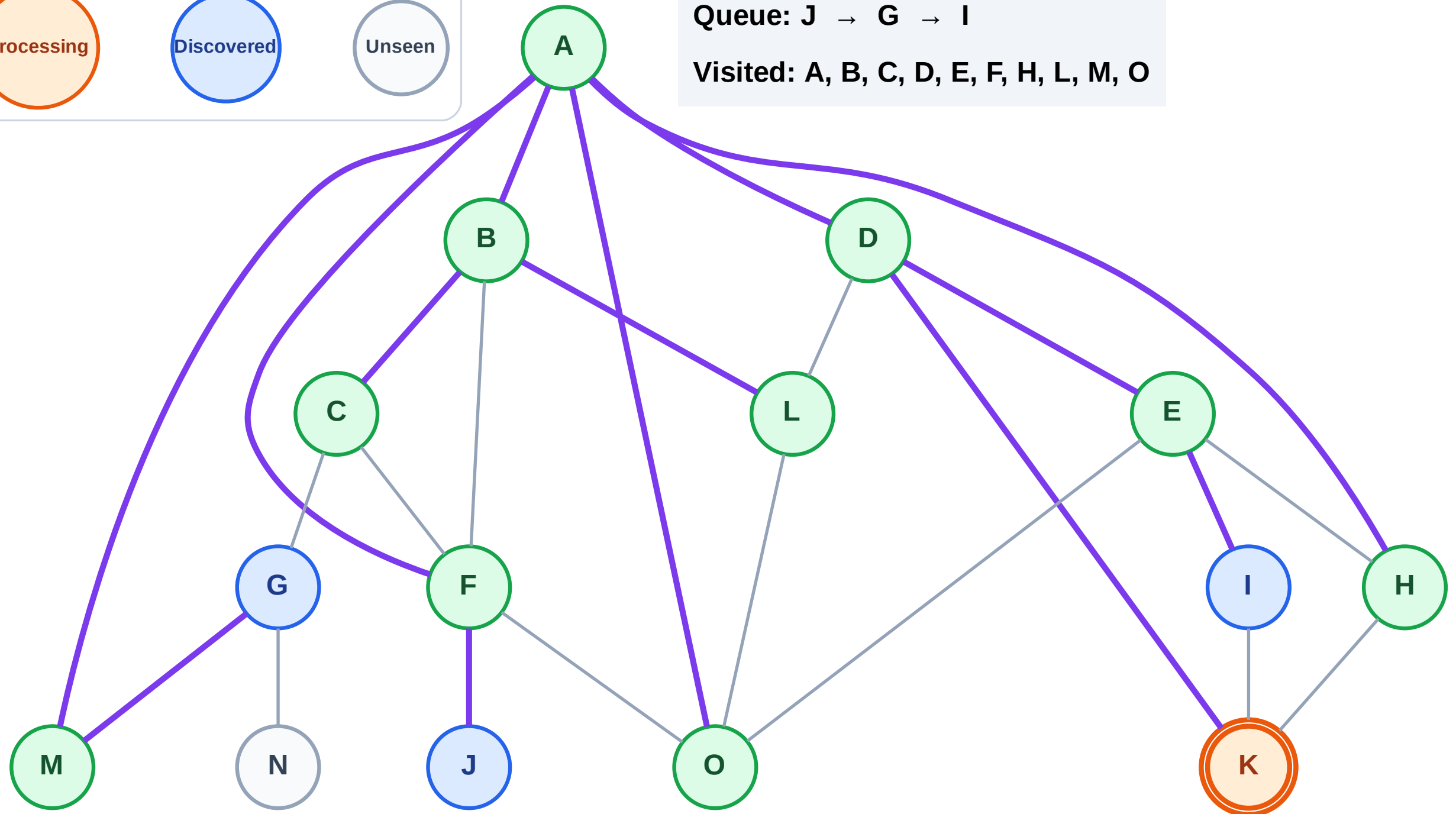
Discovered



Unseen

Queue: J → G → I

Visited: A, B, C, D, E, F, H, L, M, O



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

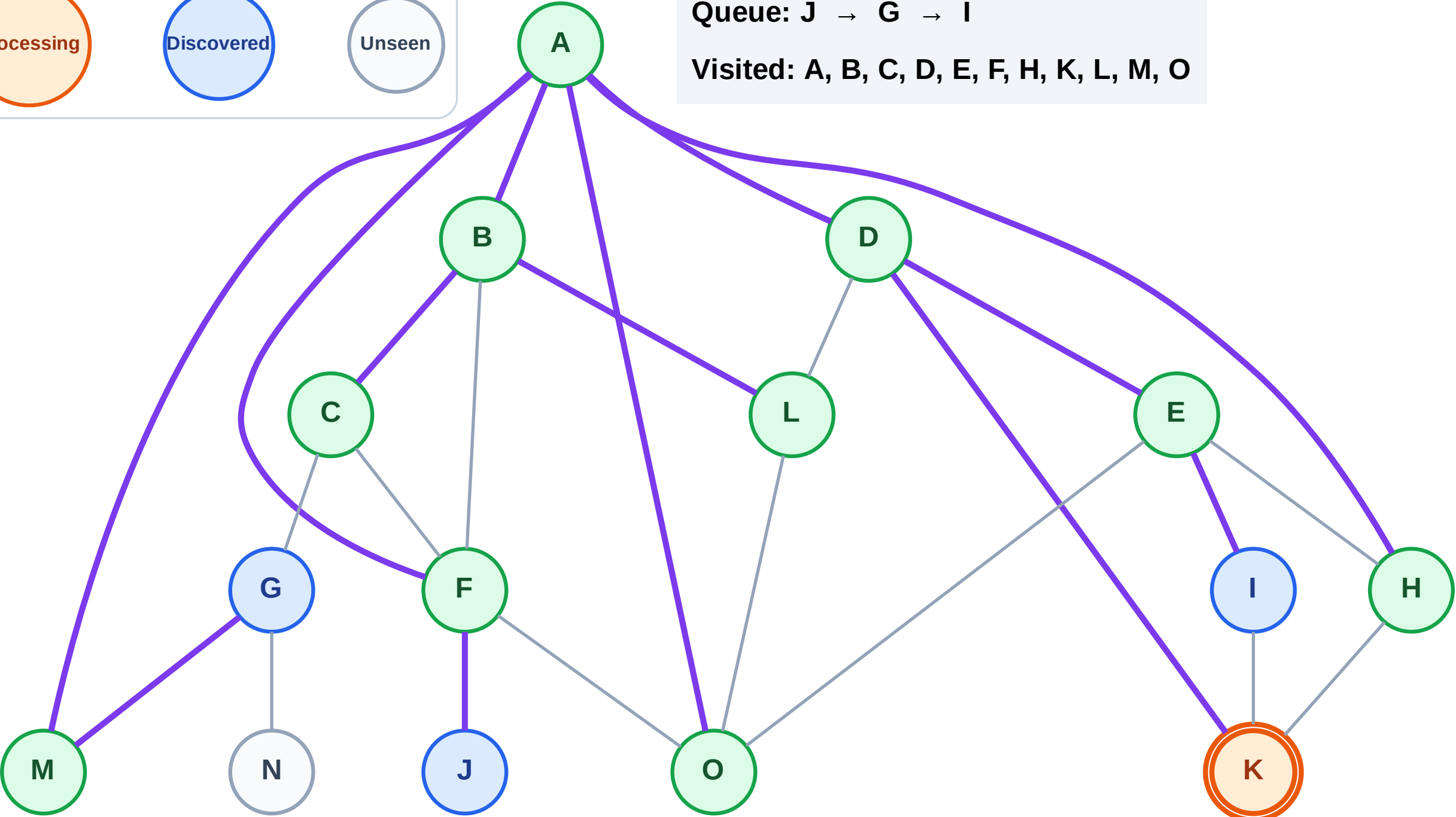
Processing

Discovered

Unseen

Queue: J → G → I

Visited: A, B, C, D, E, F, H, K, L, M, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

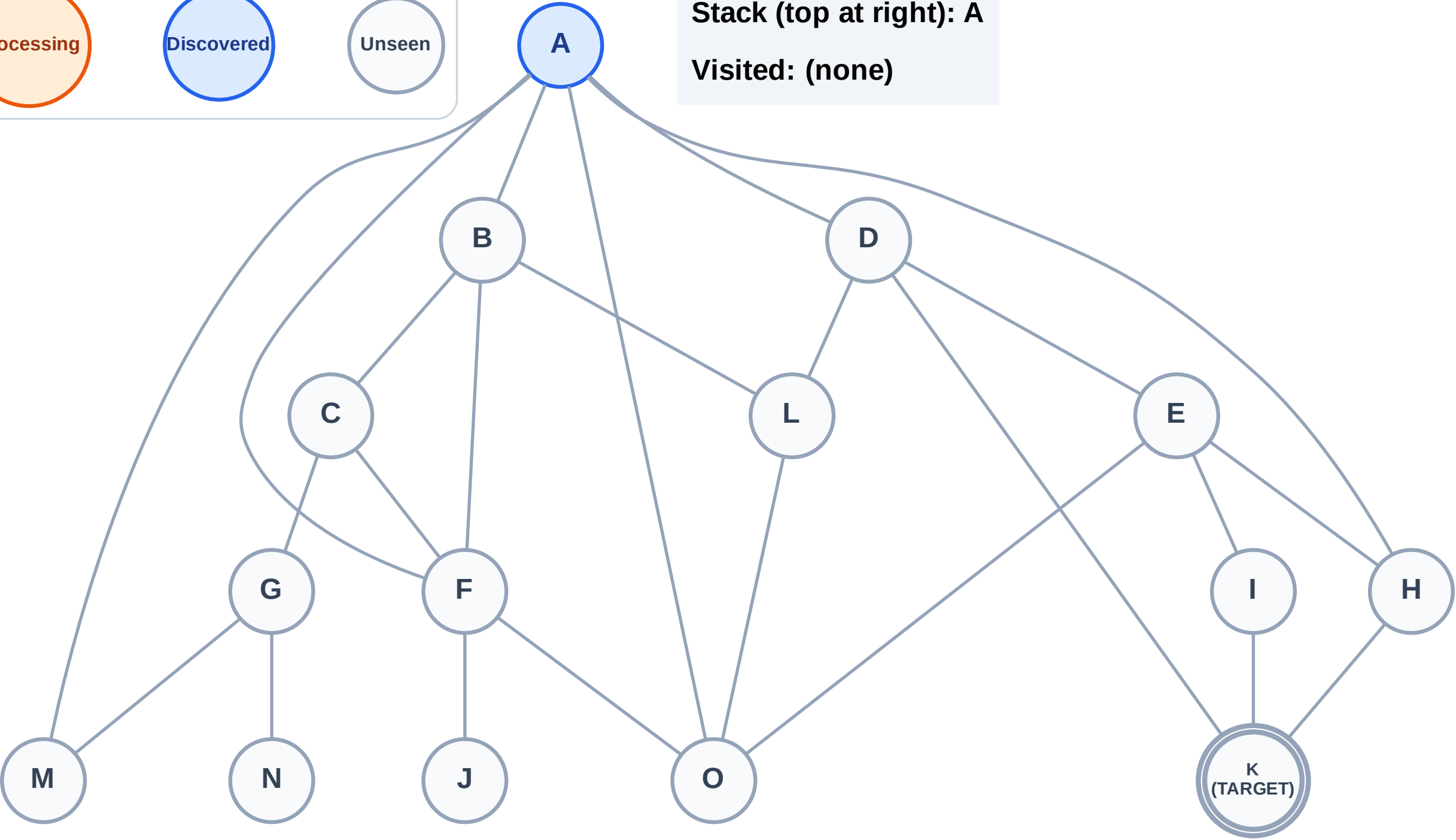
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

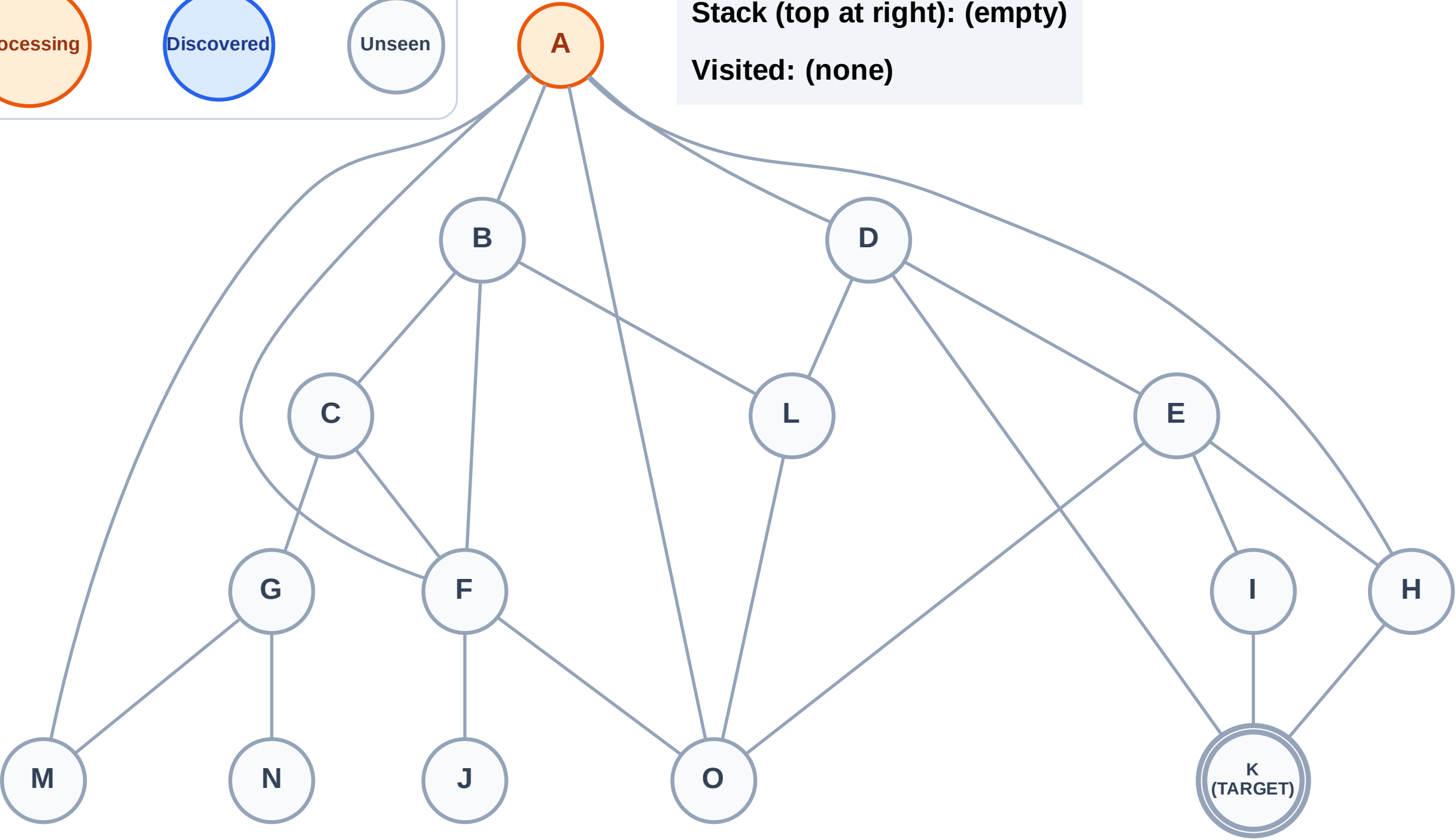
Complete

Processing

Discovered

Unseen

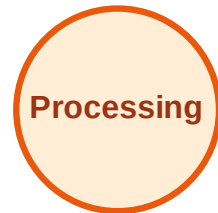
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

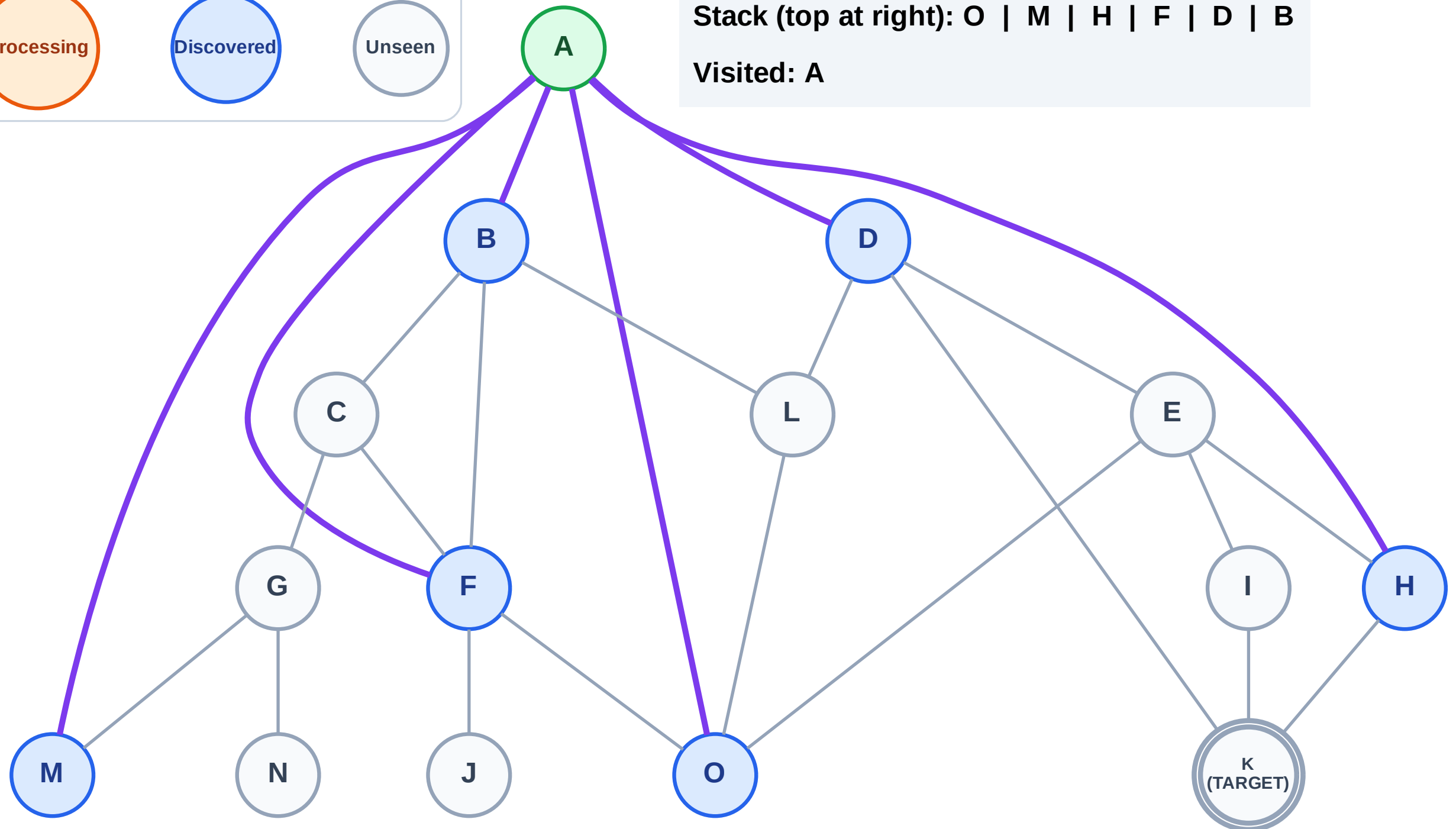
Step 3: Discover O, M, H, F, D, B from A

Legend



Stack (top at right): O | M | H | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

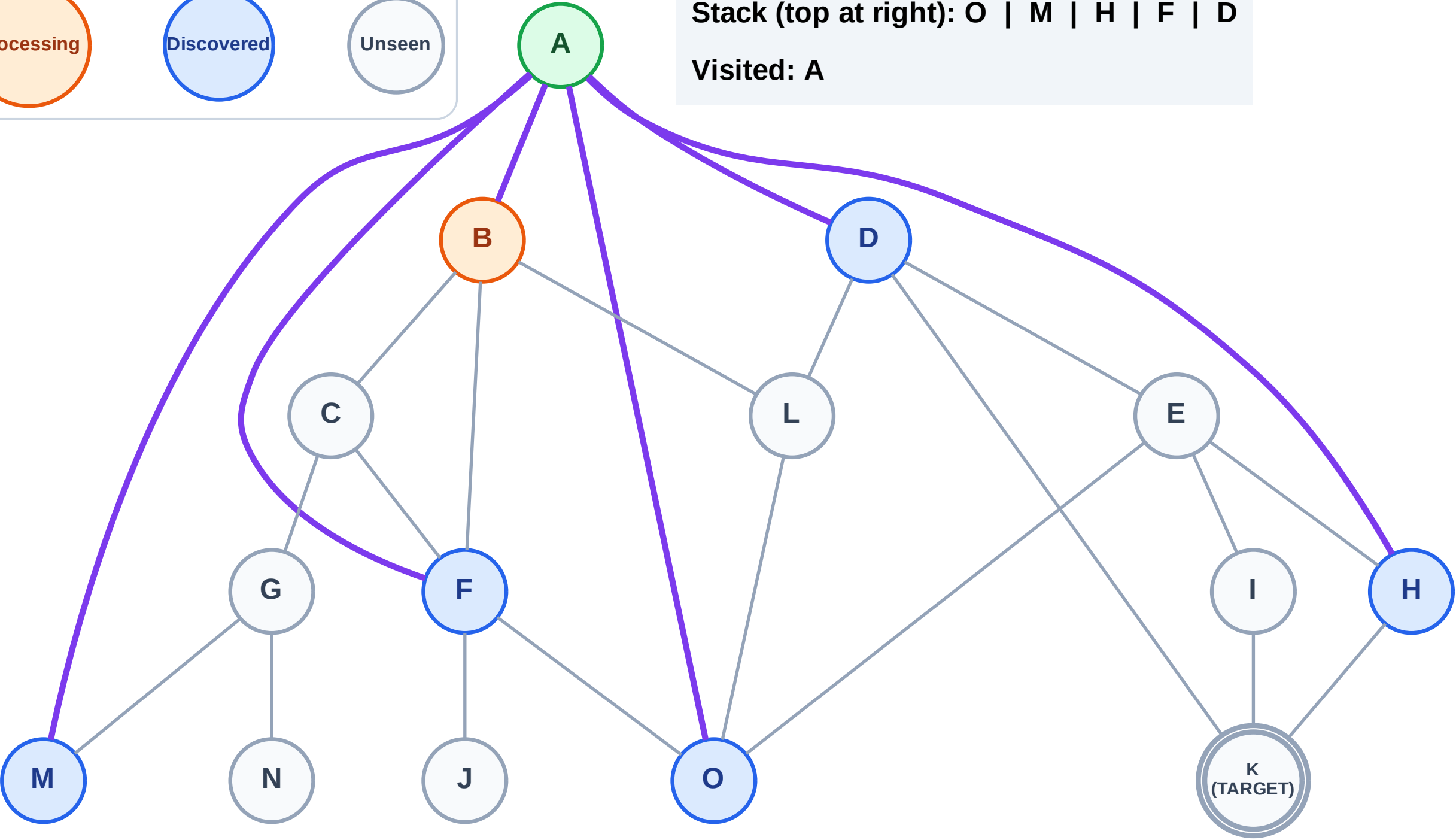
Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D

Visited: A



DFS Traversal

Step 5: Discover L, C from B

Legend

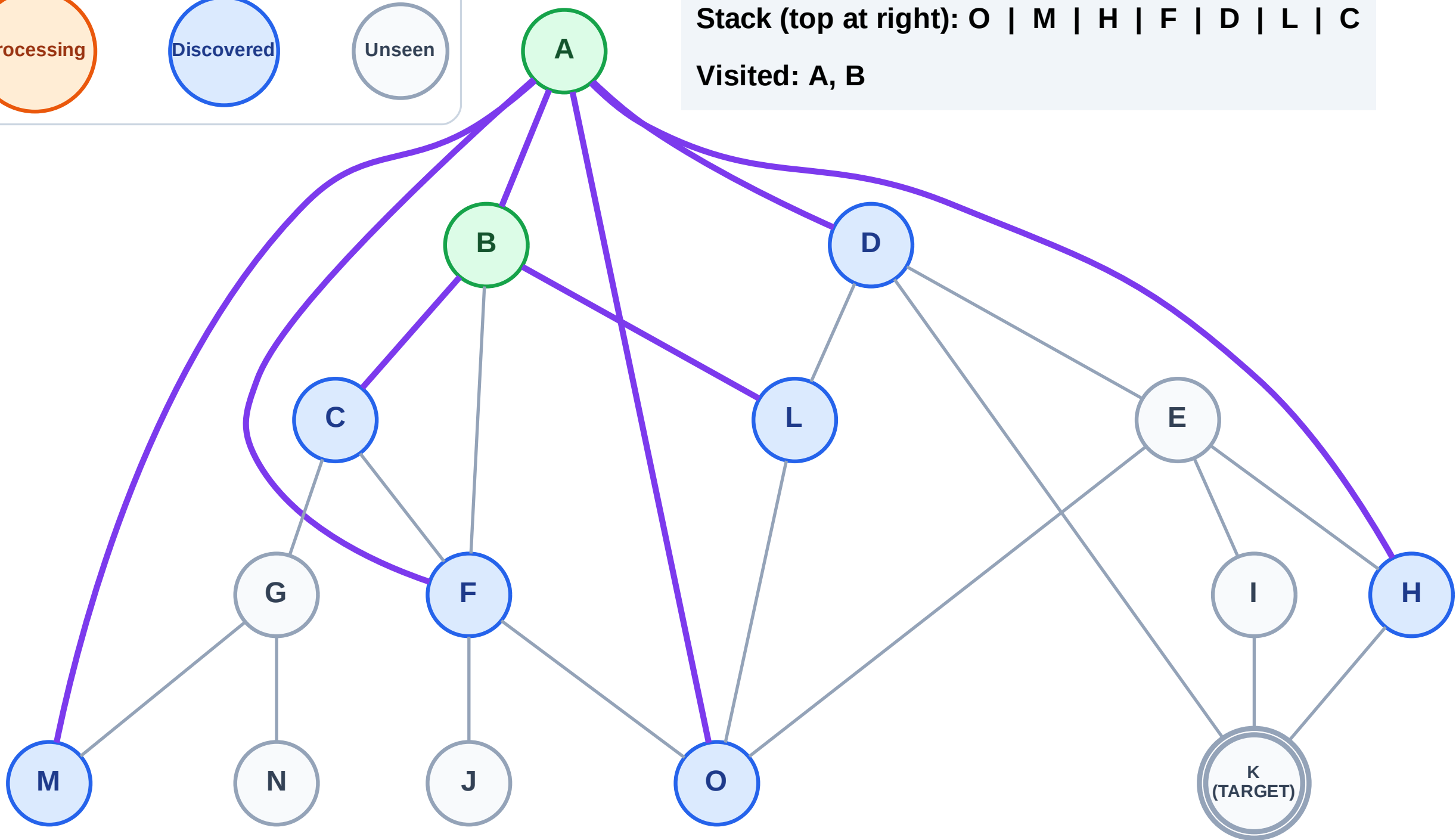
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D | L | C
Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

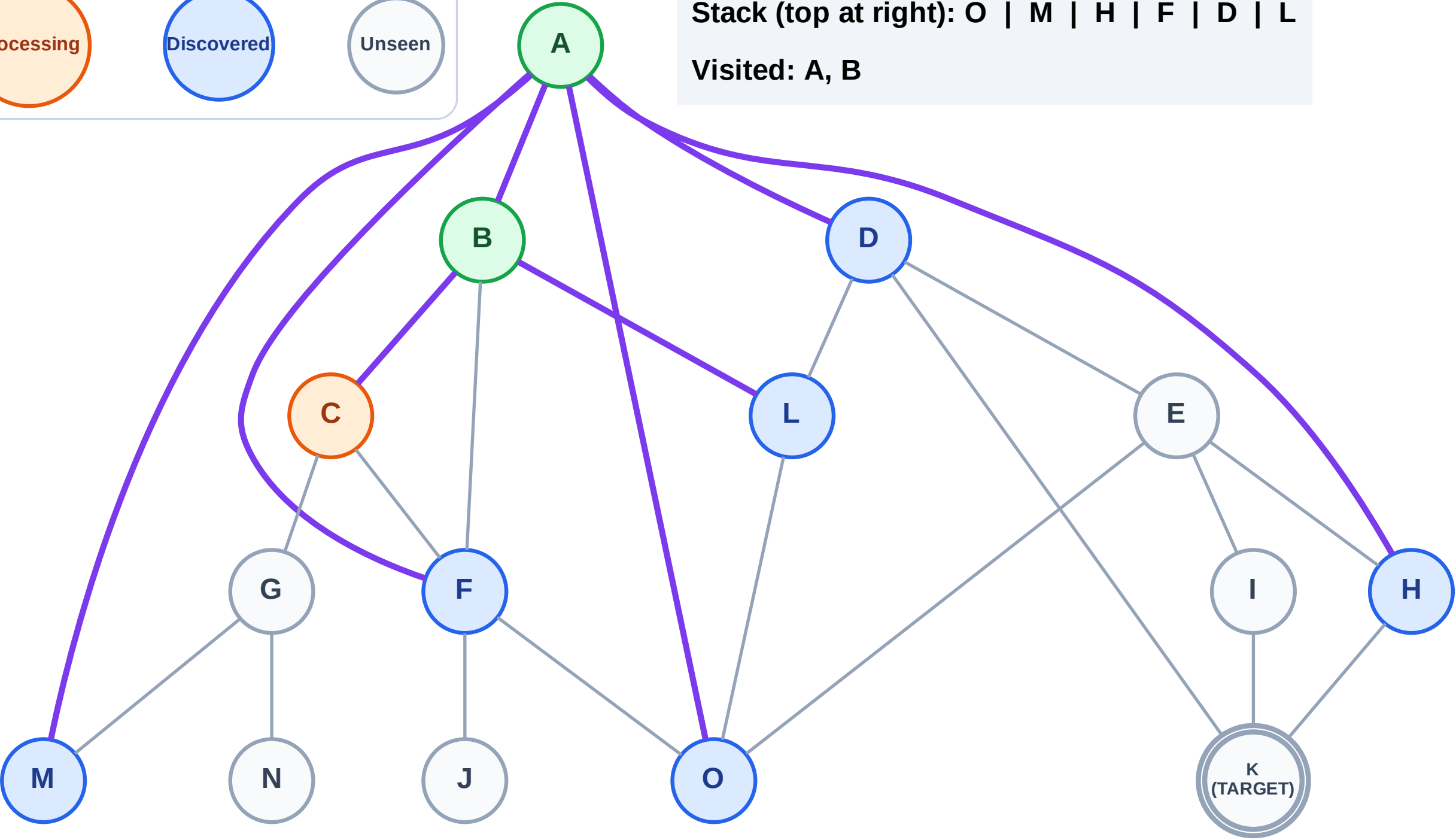
Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D | L

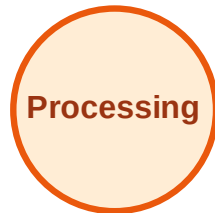
Visited: A, B



DFS Traversal

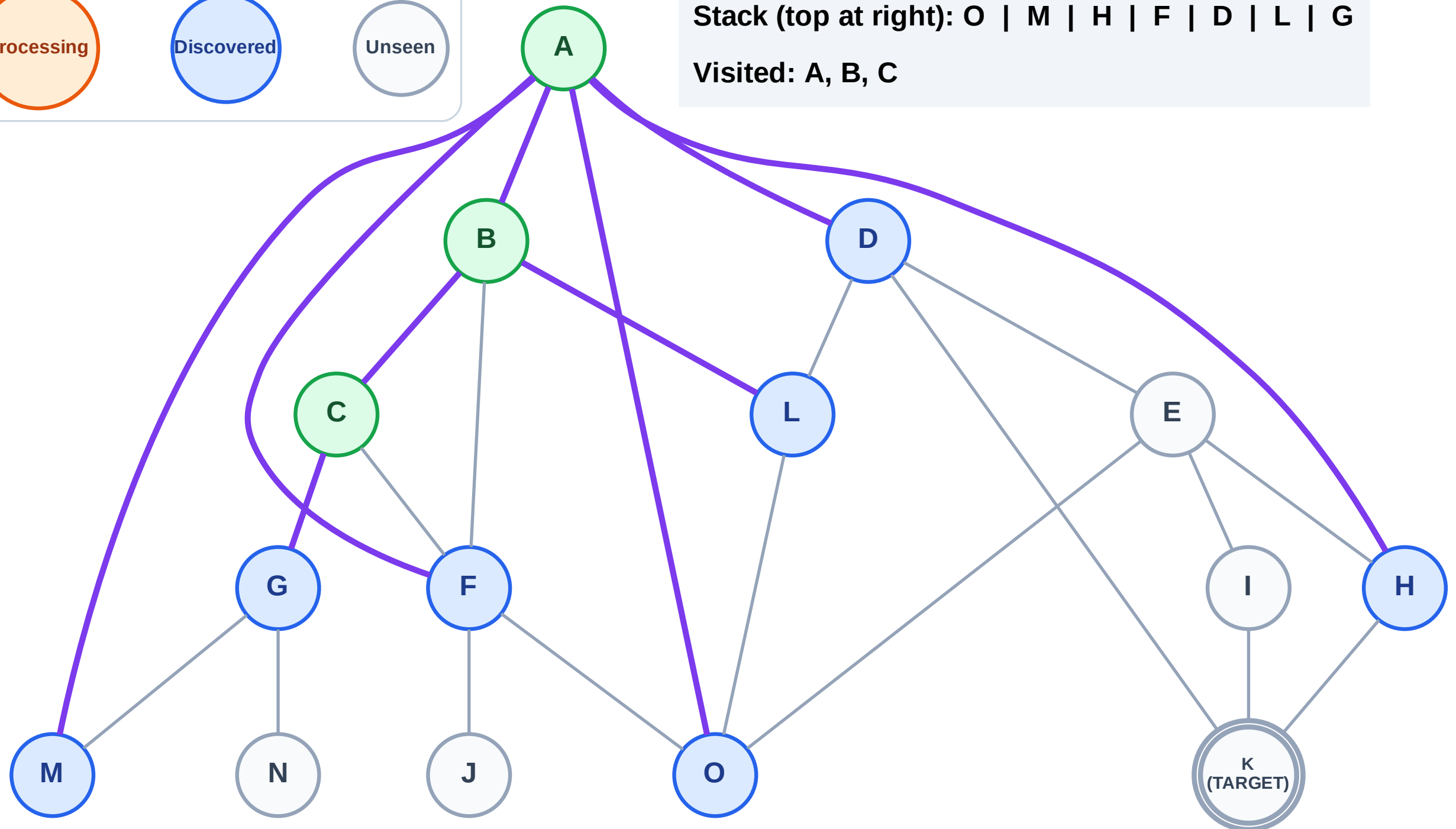
Step 7: Discover G from C

Legend

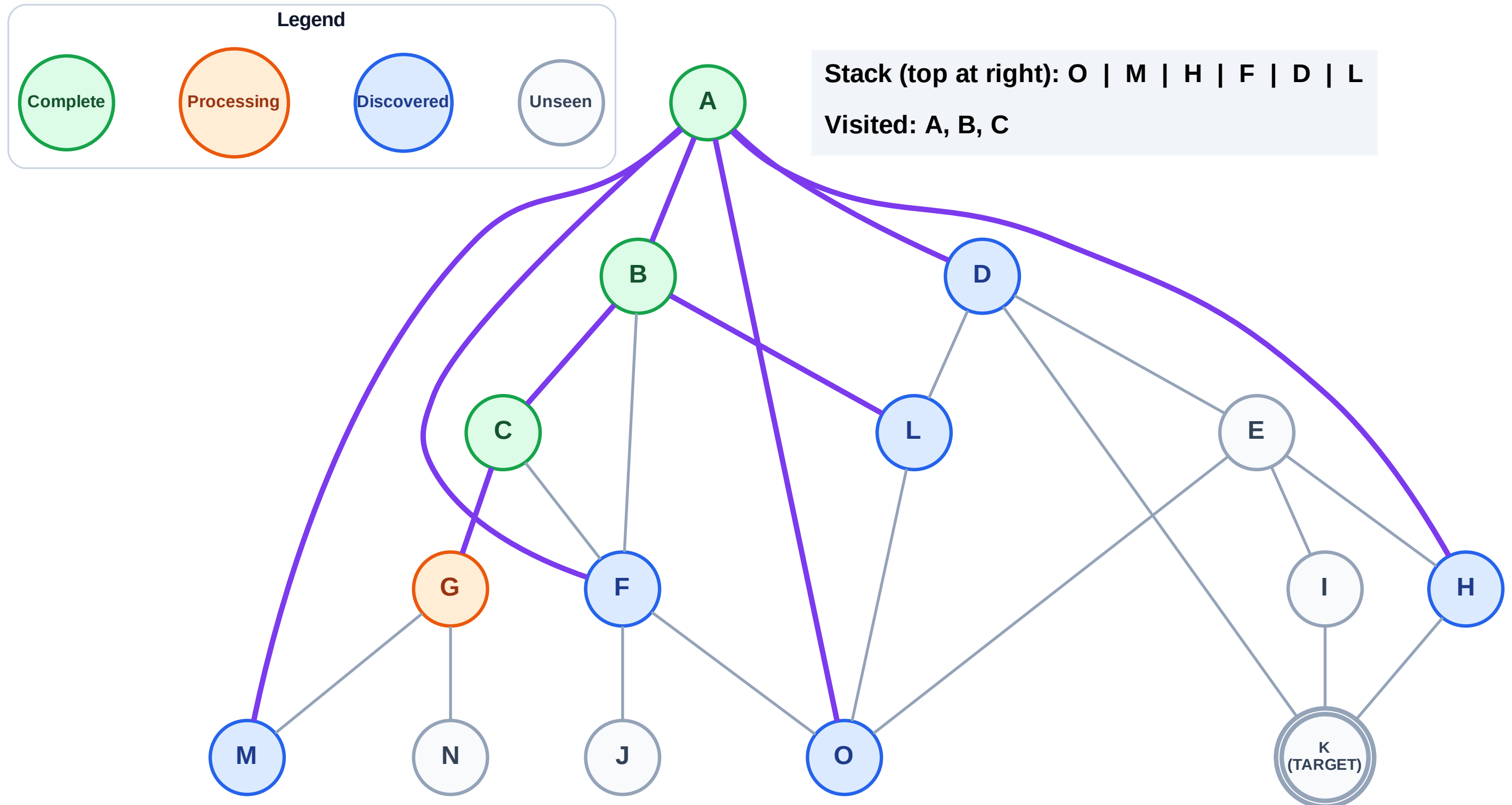


Stack (top at right): O | M | H | F | D | L | G

Visited: A, B, C



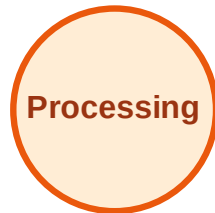
Step 8: Process node G



DFS Traversal

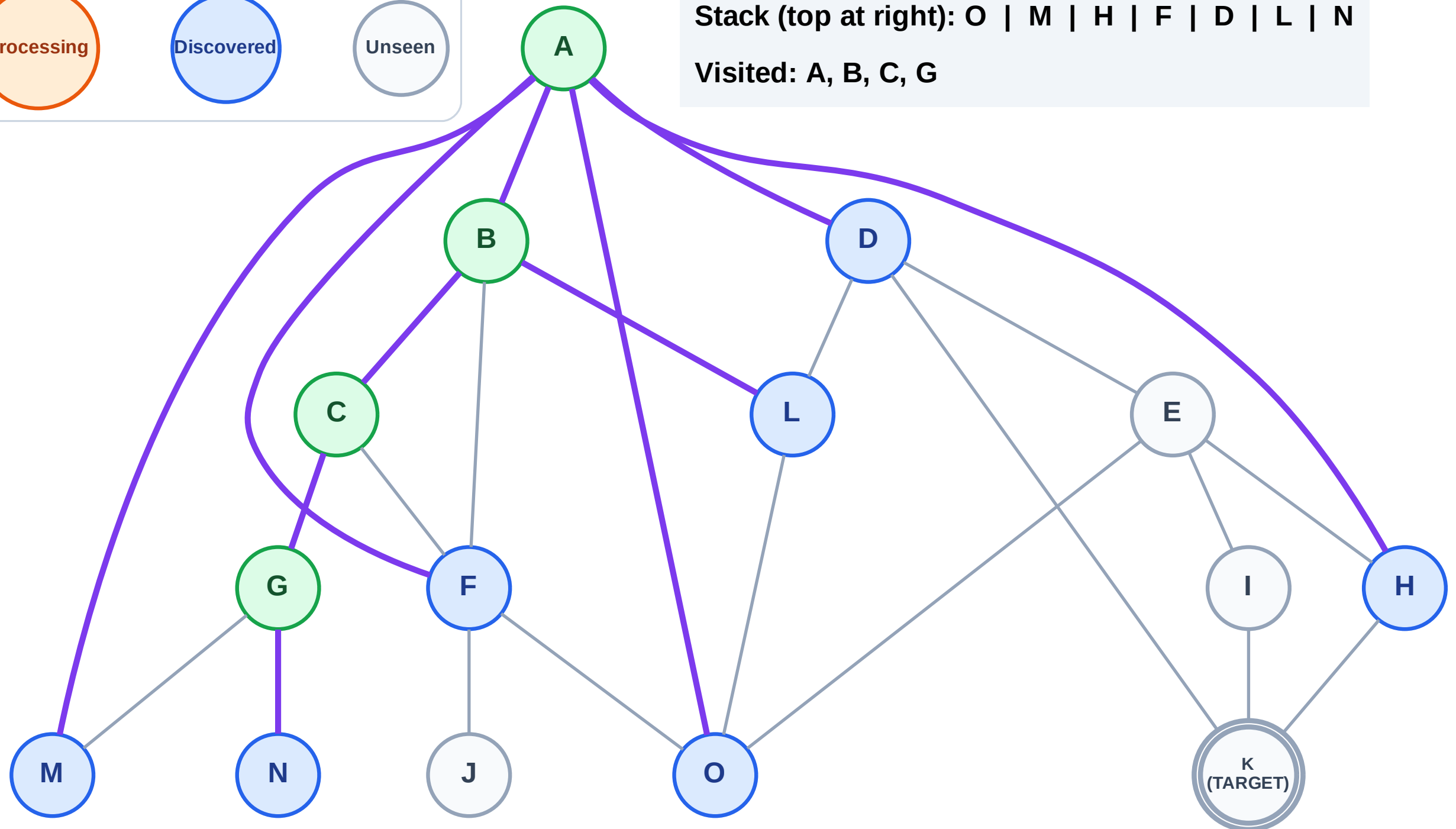
Step 9: Discover N from G

Legend



Stack (top at right): O | M | H | F | D | L | N

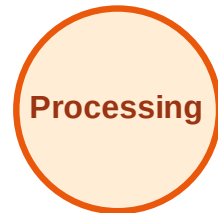
Visited: A, B, C, G



DFS Traversal

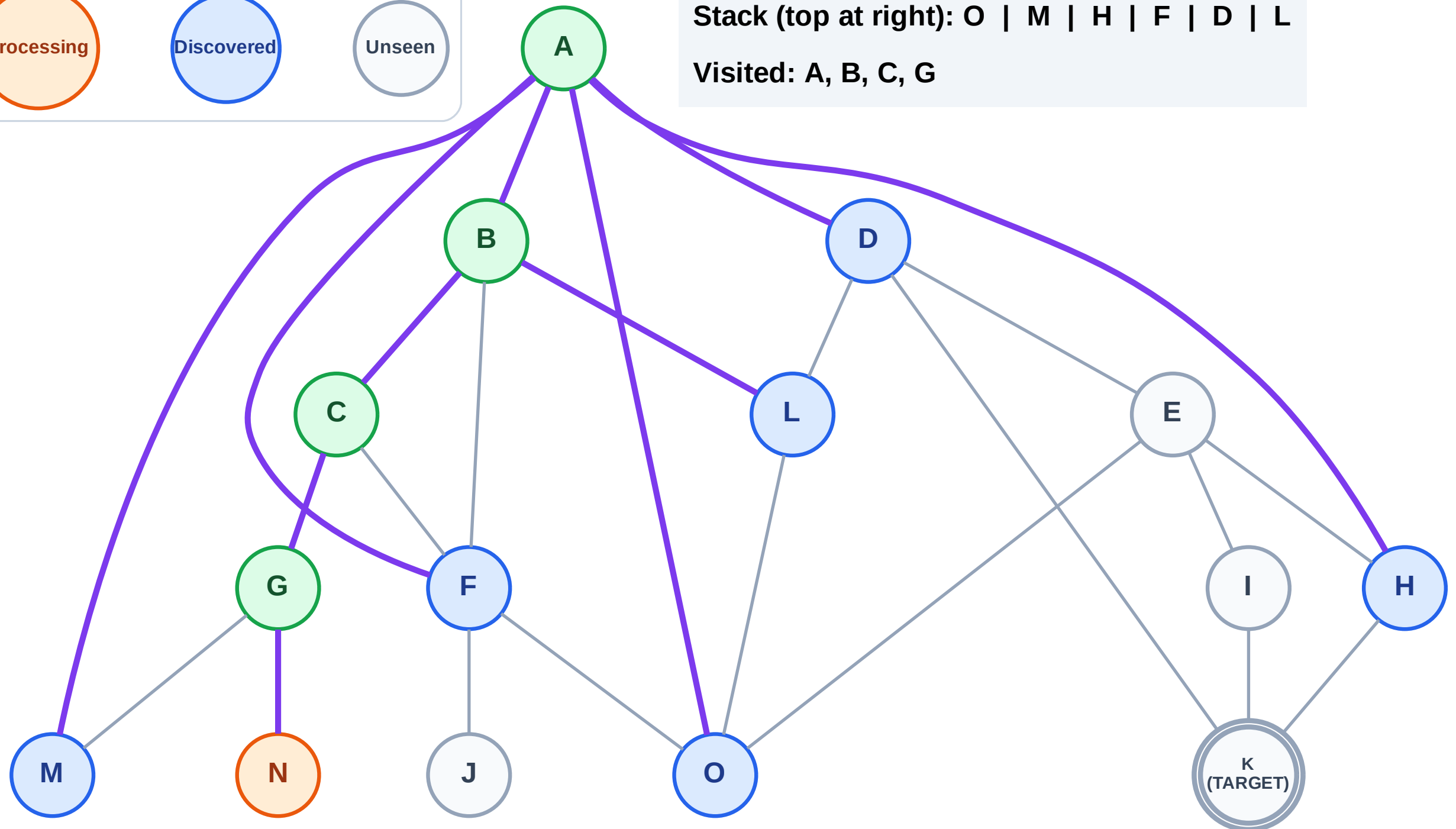
Step 10: Process node N

Legend



Stack (top at right): O | M | H | F | D | L

Visited: A, B, C, G



DFS Traversal

Step 11: No new neighbors from N

Legend

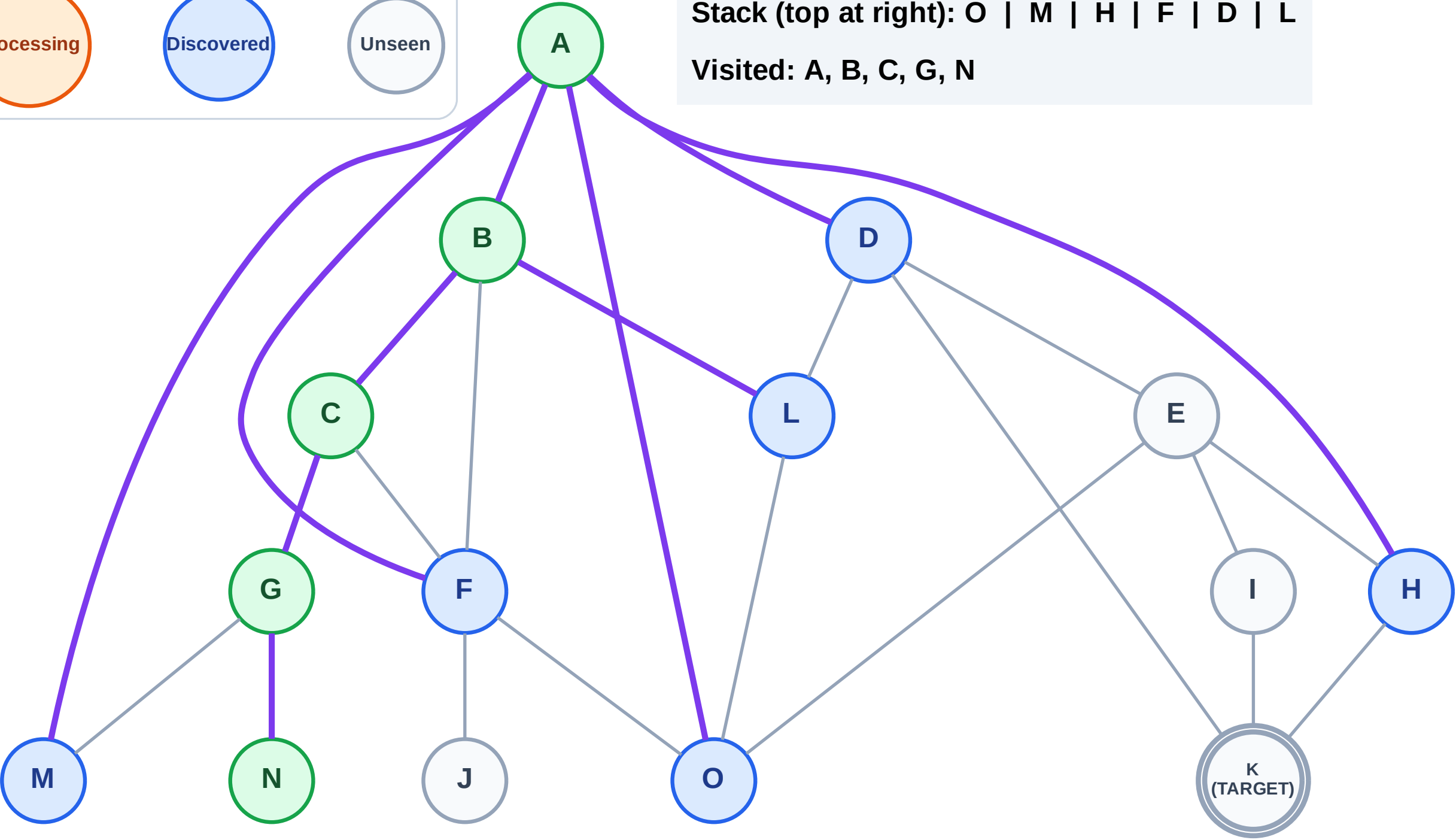
Complete

Processing

Discovered

Unseen

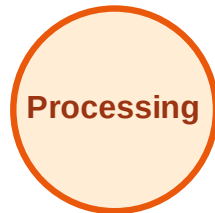
Stack (top at right): O | M | H | F | D | L
Visited: A, B, C, G, N



DFS Traversal

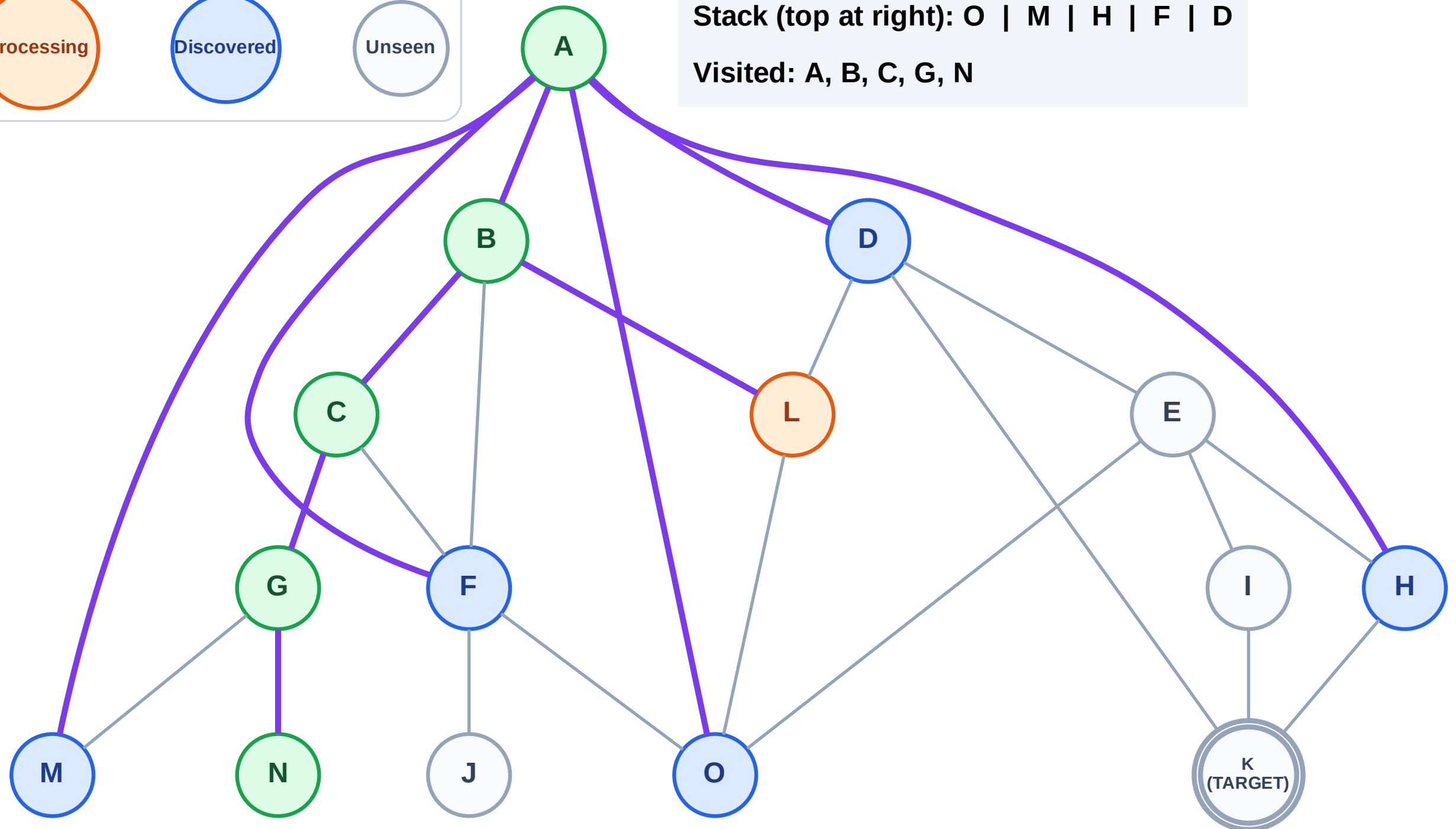
Step 12: Process node L

Legend



Stack (top at right): O | M | H | F | D

Visited: A, B, C, G, N



DFS Traversal

Step 13: No new neighbors from L

Legend

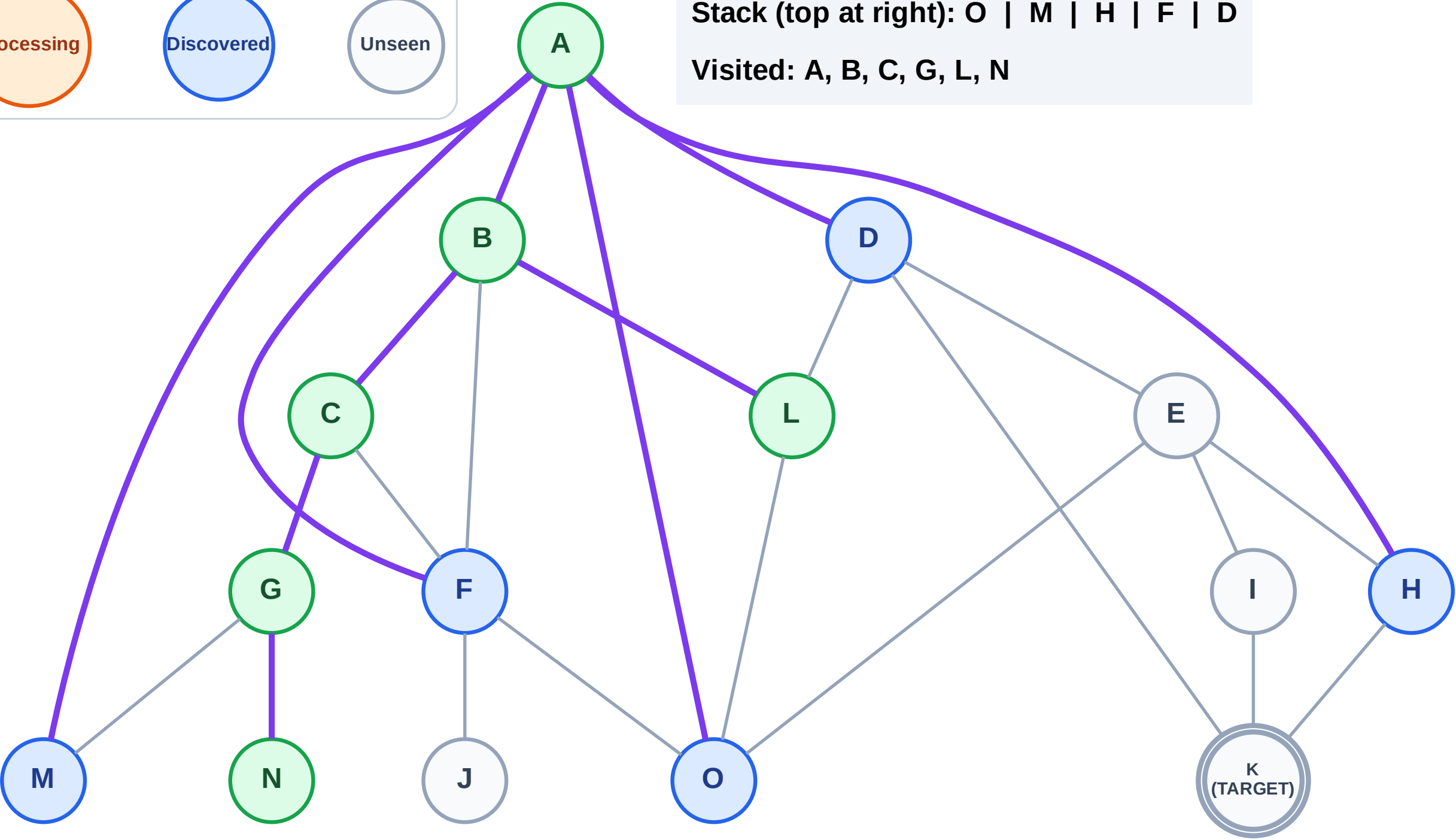
Complete

Processing

Discovered

Unseen

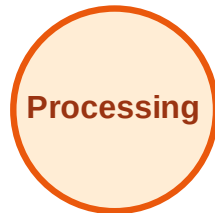
Stack (top at right): O | M | H | F | D
Visited: A, B, C, G, L, N



DFS Traversal

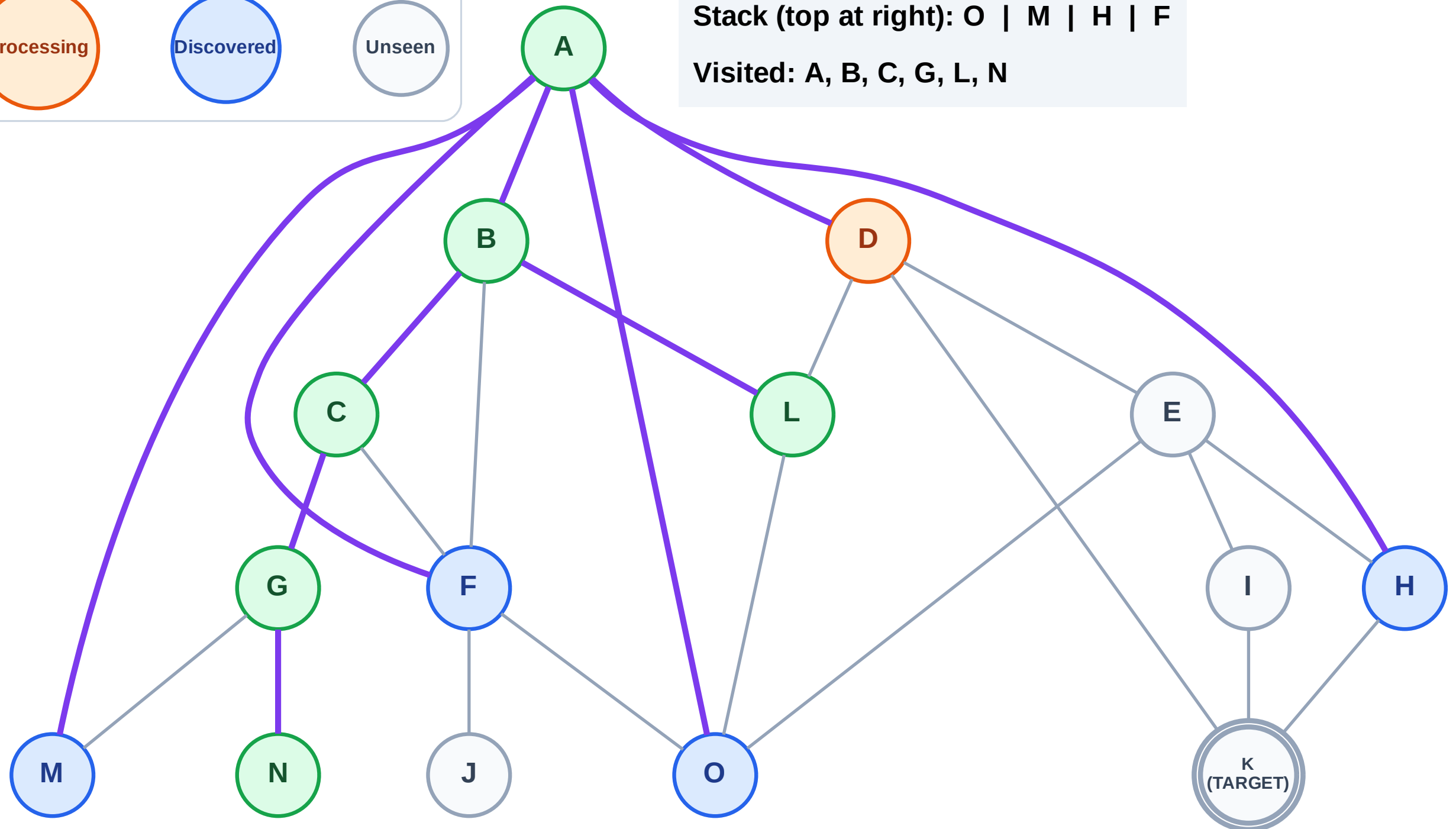
Step 14: Process node D

Legend



Stack (top at right): O | M | H | F

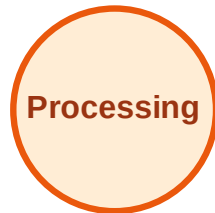
Visited: A, B, C, G, L, N



DFS Traversal

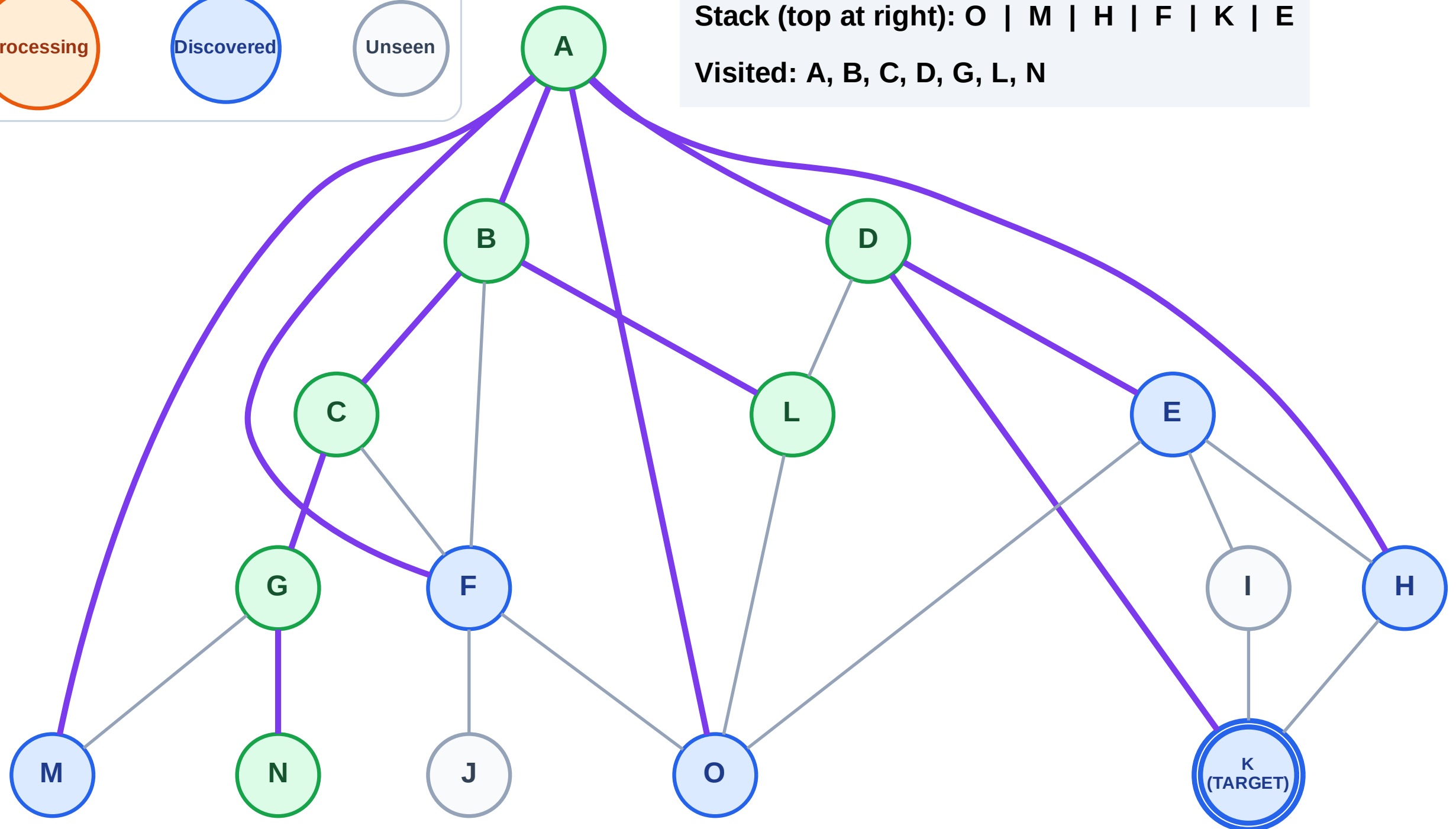
Step 15: Discover K, E from D

Legend



Stack (top at right): O | M | H | F | K | E

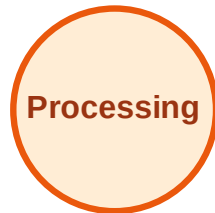
Visited: A, B, C, D, G, L, N



DFS Traversal

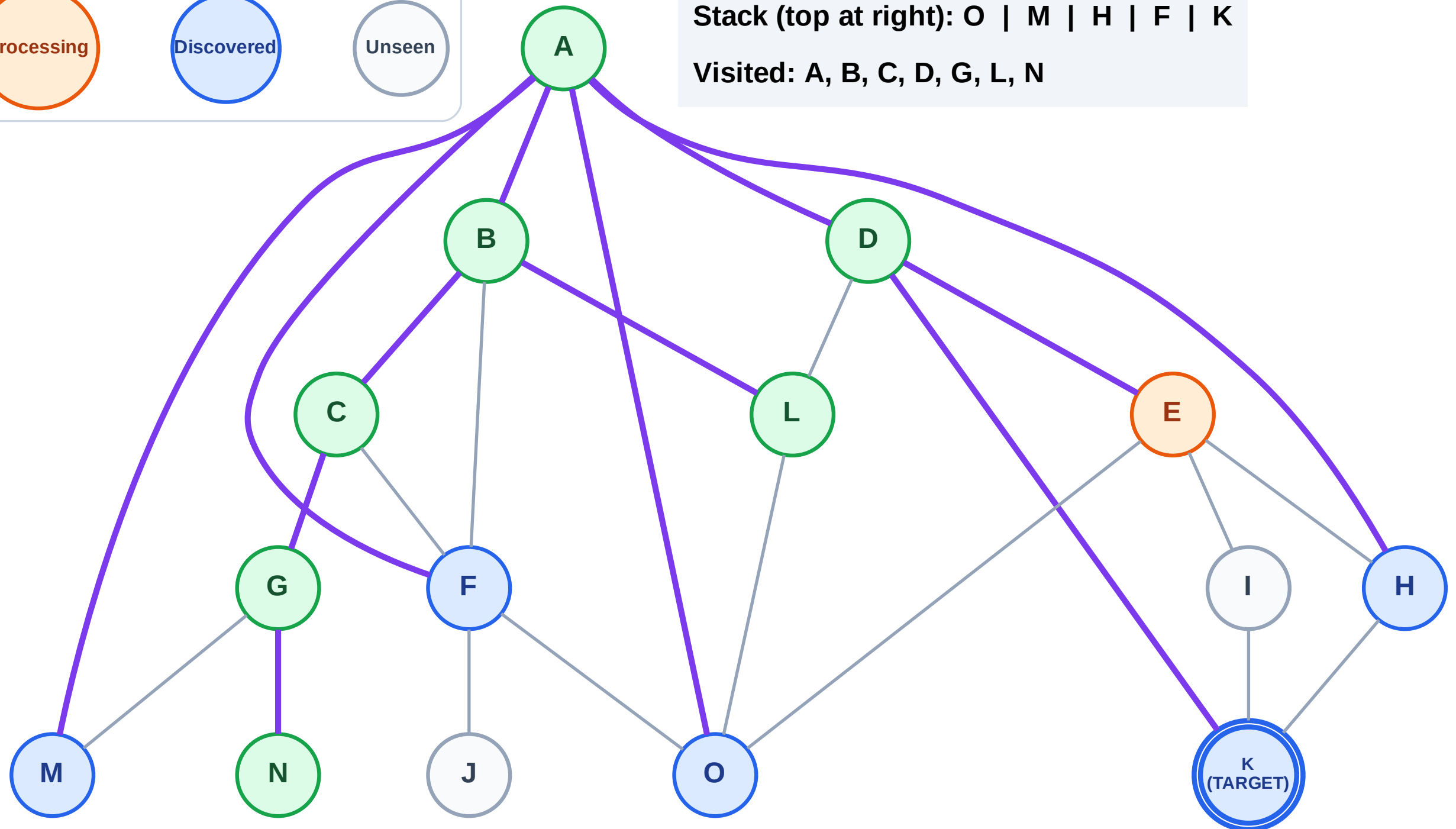
Step 16: Process node E

Legend



Stack (top at right): O | M | H | F | K

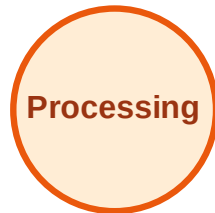
Visited: A, B, C, D, G, L, N



DFS Traversal

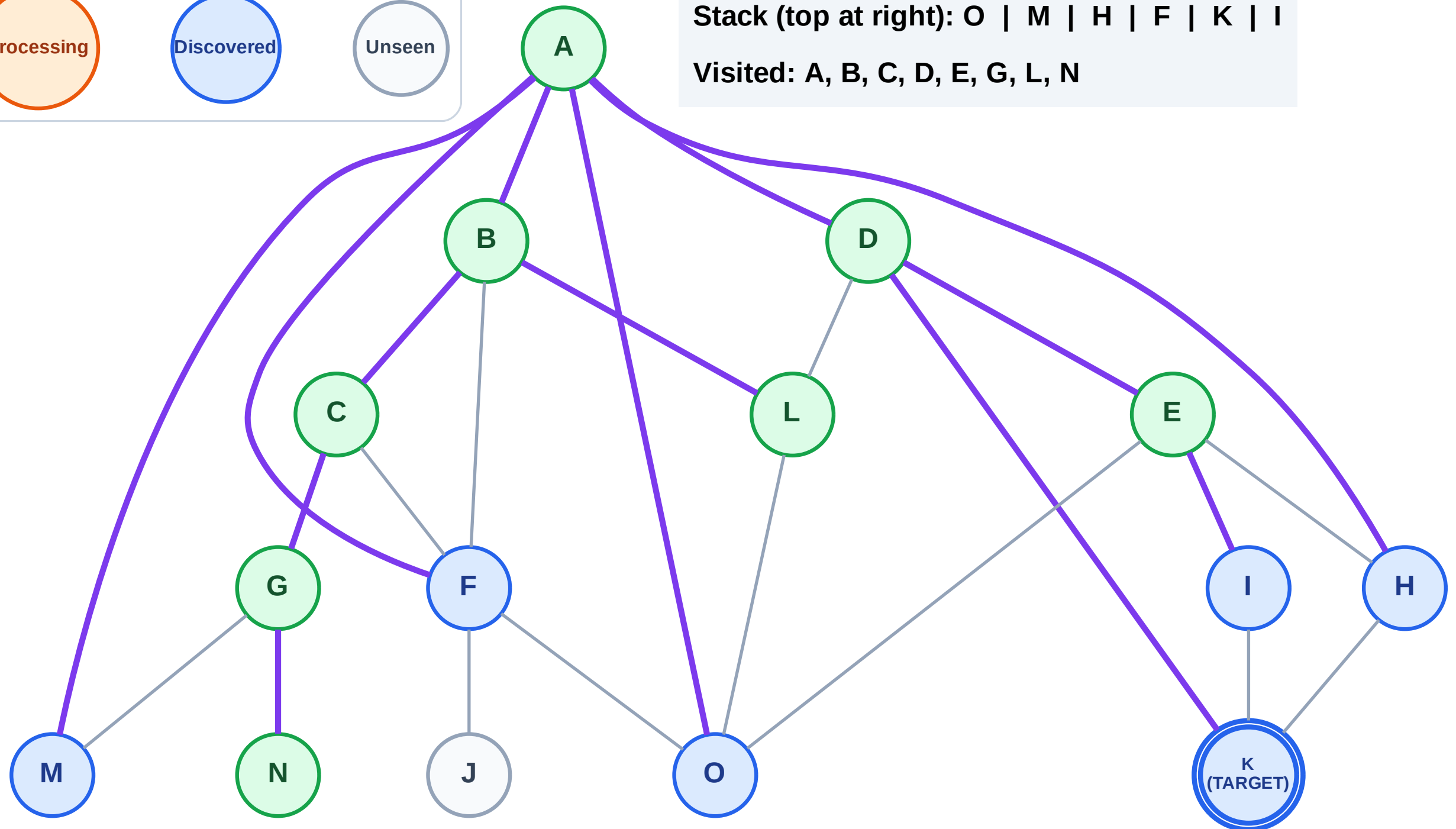
Step 17: Discover I from E

Legend

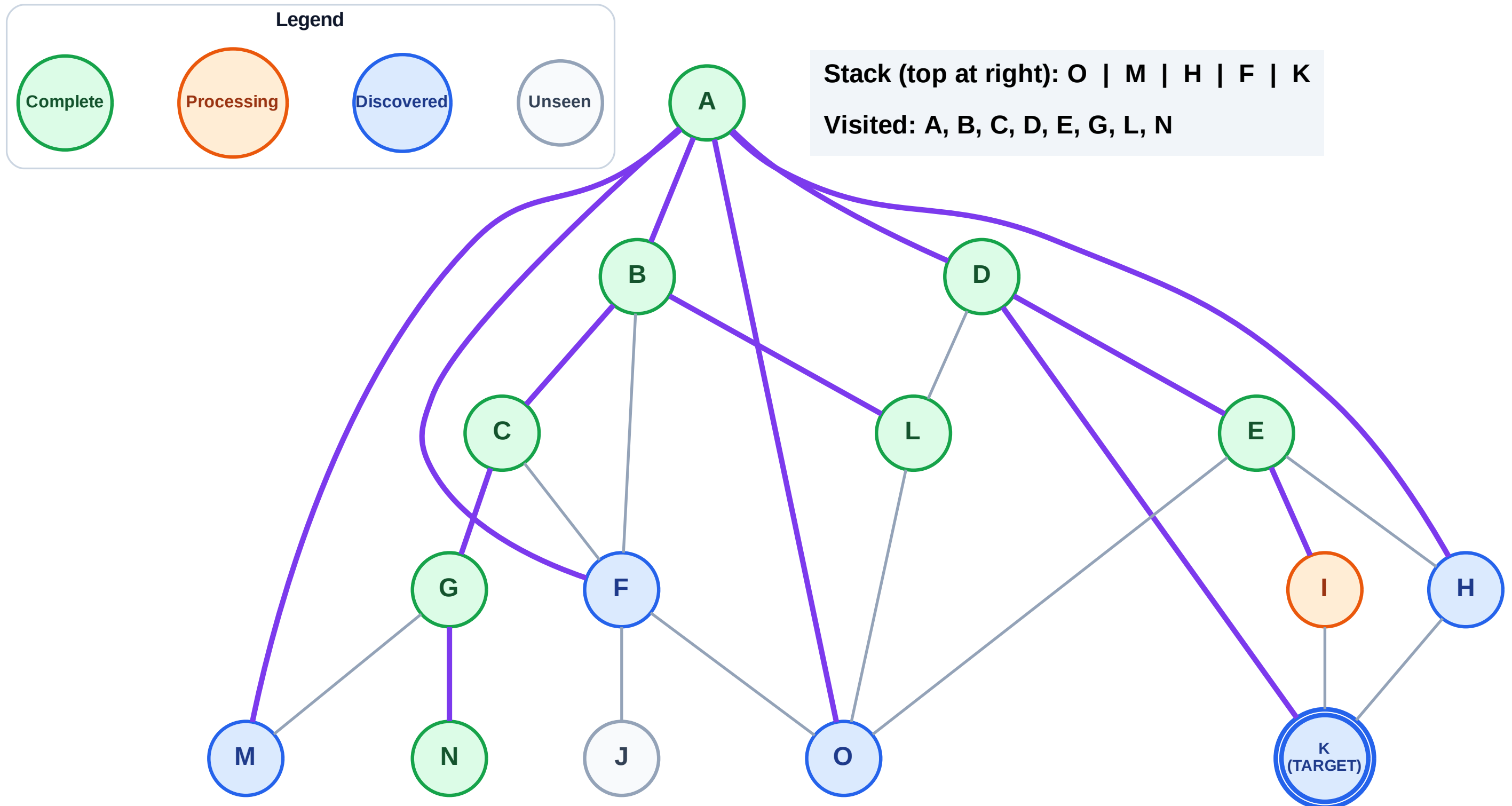


Stack (top at right): O | M | H | F | K | I

Visited: A, B, C, D, E, G, L, N



Step 18: Process node I



DFS Traversal

Step 19: No new neighbors from I

Legend

Complete

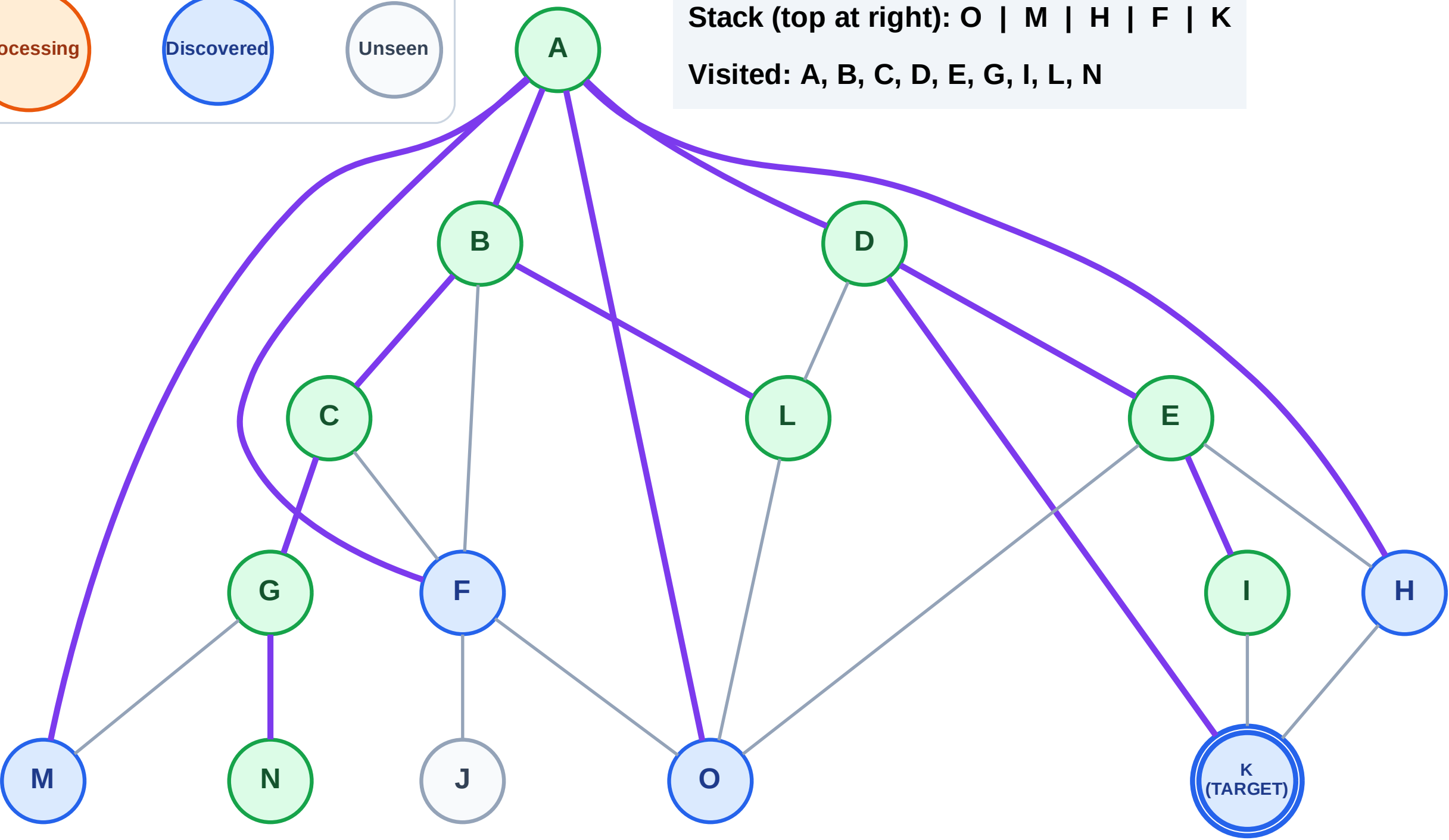
Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | K

Visited: A, B, C, D, E, G, I, L, N



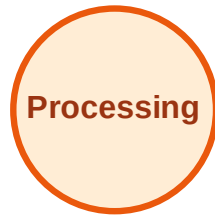
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing



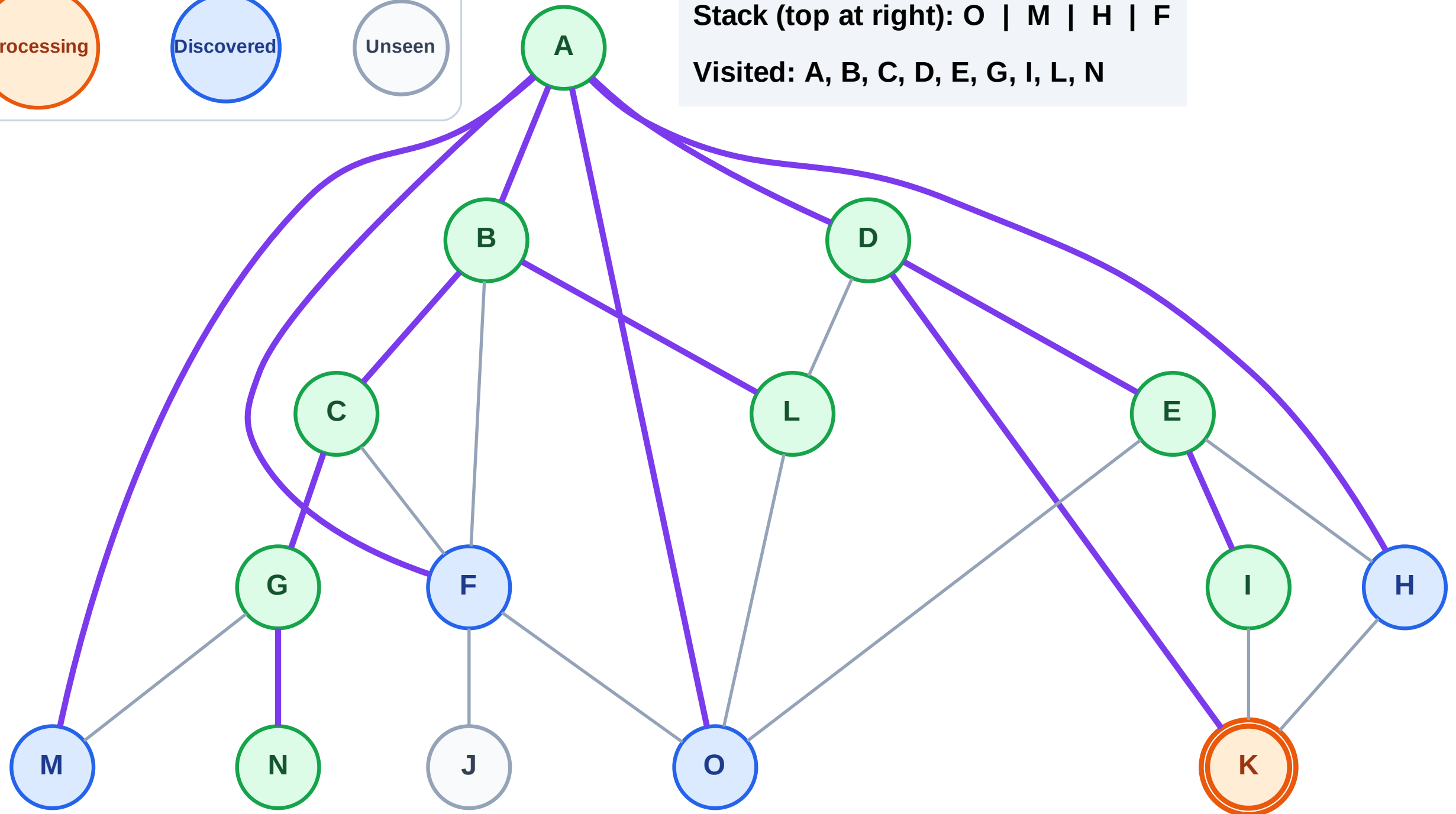
Discovered



Unseen

Stack (top at right): O | M | H | F

Visited: A, B, C, D, E, G, I, L, N



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | H | F
Visited: A, B, C, D, E, G, I, K, L, N

